ZTL — Zero-Trust Logic — v1.4

V. Reznik. Preprint, v1.4 — 2026-09-07. Concept DOI: 10.5281/zenodo.21318981 (v1.3: 10.5281/zenodo.21472971; v1.2: 10.5281/zenodo.21440066; v1.1: 10.5281/zenodo.21323552; v1.0: 10.5281/zenodo.21318982). v1.4 adds: *no numbered section rests on measurement alone* — every one of the eighteen sections that carried the MEASURED tag now names kernel-checked theorems behind its load-bearing claims, the corpus growing threefold to sixty-six modules and 1112 theorems, all on the empty axiom list (§8); *the six traditions of §1 as six embedding theorems* — IEEE 754 NaN, SQL NULL, taint tracking, abstract interpretation, Dempster–Shafer and provenance semirings, each formalised as its own tradition states it, ZTL's verdict placed inside it and the unformalised remainder named (`ZNaN.lean`, `ZNull.lean`, `ZFlow.lean`, `ZAbsInt.lean`, `ZDempster.lean`, `ZProv.lean`; §§1, 15, 16, 20, 27); *three hardness results on the warranties of §19* — the hereditary grade is a tautology check (coNP-hard, `ZHeredTaut.lean`), the exact width of an inquiry is NP-hard (`ZWidthHard.lean`), the exact receipt is NP-hard (`ZReceiptHard.lean`), so the three grades the judge does not compute are the three it cannot compute cheaply — every complexity claim held at hardness, membership argued and not claimed — with the receipt's exactness on linear claims proved under a definite reading (`LabelExactDefinite.lean`); *a syntactic cut-elimination procedure with its bound as a function* (§5, `ZCutElim.lean`); *the Lean port of the parameter tableaux of §6* — the axiom tier of the four quantifier rules measured — three sound on the empty axiom list, the fourth only classically and kept outside the corpus — a fuel-bounded search built and proved sound, and the finite half of completeness a theorem, the infinite half stated with its parts and left argued (`ZParamEngine.lean`, `ZParamHintikka.lean`; §§6, 27); the price list on the numeric floor (§15, `ZNumPrice.lean`); the global reading with theorems of its own (§10, `ContextClosure.lean`); the identity and free-instantiation results of §25 for an arbitrary domain (`ZEqGeneric.lean`, `ZFreeUIGeneric.lean`); and the precise position against Tomova's natural-implication criterion (§4).

Carried over from the published v1.3 — version DOI 10.5281/zenodo.21472971. Concept DOI: 10.5281/zenodo.21318981 (v1.2: 10.5281/zenodo.21440066; v1.1: 10.5281/zenodo.21323552; v1.0: 10.5281/zenodo.21318982). v1.3 adds: the Suszko positioning (§4) — Z as a mark rather than a third truth value is Suszko's Thesis taken as architecture rather than recovered by reduction, with the discriminator measured; the signature result (§4) — the single rule $\neg\neg p \vDash p$ separates ZTL from each of its four involutive-negation neighbours (K3, LP, weak Kleene, $Ł_3$) and, by one lemma, from any three-valued matrix with involutive negation, its cause proved in `lean/Signature.lean` on the empty axiom list; *first-order identity* (§24, `ZEq.lean`) — a = predicate whose reflexivity is an earned verdict, falling to Z on an unverified reference, while Leibniz's law licenses substitution only through an earned equality; and *free logic with definite descriptions* (§25, `ZDesc.lean`) — a non-denoting term takes the mark (existence = earned self-identity), and the greedy collapse (excluded middle on it is F, not a gap and not super-true) sets ZTL apart from the neutral free-logic school. v1.2 added: the central construction named — the zero-trust lift (§2); §3.8, an explicit Lean-verified census of the sixteen lifted binary connectives — re-deriving Finn's external-Bochvar completeness landscape (Finn 1974): solo-completeness tracks non-commutative directionality (Sheffer's stroke and Peirce's arrow fall, both implications and both abjunctions survive), and the surviving basis ↛ reads as the credit detector; the fence-depth theorem in §19 (exactly $m-1$; no constant-depth fence); the at-scale stress-test of the warranty ladder; the recalculated Bochvar ledger (§4); and the three-laws-of-thought reading — a denial is free, an affirmation is on credit (§3.1); and the paradox engine (§11) — the expeditions unified as one construction paradox(f)=ground(S=f(S)) with three measured layers; and — the headline of this version — a TEMPORAL LAYER (§§21–23) in which the only clock is the arrival of ground: the warranty ladder read as temporal quantifiers (now / at every ending / always on every path), the absorption and arrow theorems machine-checked (`ZTime.lean`), an expiry event that opens epochs, and the EPOCH BOUNDARY THEOREM (`EpochBoundary.lean`) — a verdict invariant across unrestricted epoch crossing reads none of its grounds, so non-trivial guarantees require the boundary between learning and world change; plus a price list of derivations (§23) — the alive rules transport truth but cannot mint it. The tag MEASURED means "verified by machine enumeration" (code in this repository), as opposed to "argued"; references to Lean mean proofs checked by the Lean 4 kernel with an **empty axiom list**.

Abstract

ZTL (Zero-Trust Logic) is a two-valued logic over marked inputs, generated in its entirety by a single principle: **truth is never granted on credit** — a connective returns T only if T is forced under every classical reading of the unverified. There are exactly two truth values (verdicts are always classical); the third symbol Z is a **mark** on an unverified input, not a truth value. The mark is barred from the value of any compound — the greediness theorem (`evalF_classical`, empty axiom list): every compound assertion is already T or F, the middle never appears above the atoms. This is more than Suszko's logical two-valuedness, which every structural logic has for free; it is the stronger, truth-functional fact that above the atoms the algebraic value already *is* the logical value, so the reduction has nothing left to do on compounds. Its identity among the three-valued matrices is precise and machine-checked at its cause: a single rule, $\neg\neg\mathbf{p} \vDash \mathbf{p}$, separates its consequence relation from each of its four involutive-negation neighbours (K3, LP, weak Kleene, Łukasiewicz $Ł_3$), and by one lemma from any matrix with involutive negation — the cause a broken involution, $\neg\neg Z = T$.

That the logic is not arbitrary is evidenced, case by case, rather than asserted. Six independent engineering traditions — IEEE 754 arithmetic (NaN), SQL's three-valued logic (NULL), taint tracking in security, abstract interpretation in static analysis, imprecise probabilities in decision theory, and provenance semirings in database theory — have each, over decades, reinvented a fragment of this same discipline. For each, its own semantics is formalised as the tradition states it, and a theorem on the empty axiom list places ZTL's verdict inside it — an embedding of the algebraic core, not of the whole tradition, with the unformalised remainder named in each case. Six traditions implementing fragments of one logic is the evidence that its generating principle is a denominator and not a construction of convenience.

For this logic we build: a complete semantic account with a measured price list (12 surviving laws, including modus ponens, and 14 fallen ones — all the fallen laws are "truth from form"); the discovered split between *rules* and *laws*, with a one-directional deduction theorem for the primitive arrow; a signed tableau calculus with machine-proven soundness, completeness and cut admissibility; an algebraic passport (the fallen idempotence yields exact truth detectors, the external layer is expressively complete, a definable external implication restores the full deduction theorem, Craig interpolation holds, and the Blok–Pigozzi conditions are verified on the matrix — ZTL is algebraizable, yet not self-extensional); quantifiers, first-order identity (a = predicate whose reflexivity is an earned verdict — self-identity falls to Z on an unverified reference — while Leibniz's law licenses substitution only through an earned equality) and free logic with definite descriptions (a non-denoting term takes the mark, not F and not a gap; existence is earned self-identity, and the greedy collapse makes excluded middle on it F, apart from the neutral free-logic school); a modal identification (local modality over the S5 frame of completions — versus global supervaluation); a probabilistic identification (verdicts are the {0,1}-threshold of Dempster–Shafer belief functions); a theory of verification (a verdict is a pair "value + warranty", where the warranty is a two-grade ladder: sound — never lies; hereditary — never revoked) and of evidence combination (conflict is never renormalized — Zadeh's paradox is resolved in Smets' favor); and a quarantine passport that types every refusal by its genesis — paradox (permanent), intrinsic (the stipulation is forced), underdetermined (until a choice), unverified input (until verification), inherited — with a measured stipulation theorem separating the liftable from the permanent; a temporal layer in which the only clock is the arrival of ground — the warranty ladder read as temporal quantifiers (now / at every ending / always on every path), with the absorption and arrow theorems machine-checked, an expiry event that opens epochs, and the Epoch Boundary Theorem (a verdict invariant across unrestricted epoch crossing is constant, so non-trivial guarantees require the boundary between learning and world change); and a price list of derivations (the twelve alive rules transport truth but cannot mint it — from no premises nothing is derivable, even the guarded tautologies, even on credit). The entire development — the core, both engine certificates with cut admissibility, the algebraic witnesses, the general fixed-point theorem, the expedition twins, the temporal modules and the frame's own mini-theorems, sixty-six modules in all — is formalized in Lean 4 **with an empty axiom list, definitions included**: 1112 theorems, each one audited individually rather than by sample (`inventory/axiom_audit.py`, re-run on every push). As of this revision no numbered section rests on measurement alone (the scouting subsection §3.7 still does): every one of the eighteen that carried the MEASURED tag now names kernel-checked theorems behind its load-bearing claims. *The tag says what is backed, not that everything in the section is* — several sections still list claims that are measured and not formalised, and each says which ones. As a test bench the logic is run over the classical paradoxes — the liar, Jourdain's carousel, Curry, Yablo, the crocodile, Russell — and in every case explosion is replaced by pointwise quarantine (for Russell, 8 of 9 membership facts stay grounded; the uncountability of the continuum splits into two independent failures). Functionally the $\{\neg,\wedge,\vee\}$ fragment coincides, cell by cell, with the external layer of Bochvar's logic (1938) — a kinship found in the literature search after the tables had been generated, not a source; the contribution of this work is the generating principle, an implicational floor lying outside the Rosser–Turquette standardness conditions, the calculus, the machine verification,

and the bridges to the engineering traditions. An interactive studio ships with the repository: natural language is negotiated into ZFL (the Zero-trust Formal Language), a small language whose validity guarantees loadability, and judged by the measured core — the pipeline itself obeys the logic it serves (the LLM's output is an unverified input; the deterministic core is the customs).

1. A logic, and six traditions that reinvented it

This paper presents a logic and argues that it was already in use. The logic is generated by one principle — truth is never granted on credit (§2) — and has a precise identity among the three-valued matrices: a single rule, $\neg\neg p \vDash p$, separates its consequence relation from each of its involutive-negation neighbours — and, by one lemma, from any matrix with involutive negation — its cause proved on the empty axiom list (§4). What keeps it from being one more many-valued table is where it was found. Most of the data over which modern software computes is unverified — sensor readings, user input, third-party databases, answers of network services — and engineering, facing this, answered not with a single theory but with the same local invention six times over:

- IEEE 754 (1985): NaN — arithmetic is infected, comparisons refuse;
- SQL (1986): NULL — three-valued logic inside expressions, forced falsehood at the WHERE boundary;
- taint tracking (Denning, 1976; Perl taint mode, TaintDroid): a distrust mark flows through computations, and only an explicit check sanitizes;
- abstract interpretation (Cousot & Cousot, 1977): interval values flow, assertions are checked for being forced;
- imprecise probabilities (Walley, 1991) and Dempster–Shafer theory: ignorance is an interval [Bel, Pl], not a point probability;
- provenance semirings (Green–Karvounarakis–Tannen, 2007): trust in a fact is a polynomial over its sources.

Each of these inventions parried its own special case of one disease: naive treatment of the unverified manufactures confidence out of nothing. We exhibit, for each of the six, a worked case in which our core reproduces that practice's central move (§§12–16, 20), and argue from those six correspondences to a common denominator — a two-valued logic over marked inputs with a single generating principle. The claim ceiling, stated here rather than left to the reader: a reproduced case is not an embedding. That ceiling no longer stands for any of the six: each has its semantics formalised and a theorem placing ZTL's verdict inside it (§27). (The ordinals below — "the sixth", "the fifth" — are the order in which the embeddings landed, not the order of the list above.) Taint tracking is the sixth (`ZFlow.lean`, empty axiom list): Denning's lattice in its two-point fragment is the common target, Perl's propagation and ZTL's are both proved to be its join, the sink is again a sign — and the three part on declassification: Denning never lowers a class, Perl lowers it by convention (the regex capture clears a bit whose payload still depends on the tainted input), ZTL by proof or not at all (`0 · mark = 0` is earned because the value is forced for every reading; no pointwise function clears a mark). SQL's NULL is the fifth (`ZNull.lean`, empty axiom list, §27): its three-valued expression layer is a homomorphism onto the lazy register, its comparison with NULL lands on the very atom IEEE's == did, and its boundaries are the four signs — WHERE is `SignT`, CHECK is `SignP`, the `<boolean test>` the rest — with WHERE agreeing with ZTL's verdict on every negation-normal search condition and parting exactly on `¬¬Z`. IEEE 754's NaN is the fourth (`ZNaN.lean`, empty axiom list, §27): the four-way comparison and the §6.2.3 propagation rule are formalised as the standard states them; arithmetic infection is a homomorphism into the mark-carrying integers; every ordered predicate is proved to be the T-sign of a ZTL atom, the unordered predicate the mark test, and != — alone — the N-sign, which is exactly where IEEE's `x != x` and ZTL's refusal of `¬(x = x)` part. The algebra of semiring provenance is formalised and mapped into the lazy register as a homomorphism, with the greedy operations proved to admit no such structure at all (`ZProv.lean`, empty axiom list, §27); and Dempster–Shafer is formalised as its own theory states it, with our verdict proved to be its {0,1}-threshold for every finite frame and every proper mass assignment (`ZDempster.lean`, §16); and abstract interpretation's Galois connection is formalised, with our verdict proved EXACT rather than merely sound on the abstract value, and the point where exactness fails named in the same file (`ZAbsInt.lean`, §15). All six are partial in the same way, and it is worth naming: provenance's K-relations, Dempster–Shafer's combination rule, abstract interpretation's fixpoint framework, IEEE's rounding, signed zero, infinities and signaling NaN, SQL's tables, joins and aggregates, and the multi-level lattices and implicit flows of information-flow control are not formalised. So six algebraic cores are closed, not six traditions. Six demonstrations became theorems, each with its unformalised remainder named. What is shown without qualification is that the denominator survives a full logical development: a calculus, quantifiers, modal and probabilistic semantics, machine verification. Along the way the classical paradoxes of self-reference, from the liar to Russell, receive a uniform diagnosis (quarantine instead of explosion) — they serve as a test bench, not as the point of departure.

The principle from which everything is built borrows its name from security: default deny. A defective input may be granted neither classical value, but every compound assertion about it must receive a classical verdict — “true only if forced”.

2. Definitions

Truth values: T (earned truth), F (falsehood). **Input mark:** Z (zero-trust, “not earned”) — a property of an atomic datum, not a truth value; it participates in the calculating tables as a third symbol.

Definition (the zero-trust lift). For any classical connective f, its lift:

$$\texttt{f*}(x_1,\dots,x_n) = \wedge \{ \ f(v_1,\dots,v_n) : v_i \in \texttt{subs}(x_i) \ \},$$

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where subs(Z) = {T,F}, subs(v) = {v} otherwise;
∧ is classical conjunction over all combinations.
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Every occurrence of Z is substituted independently; the result is always classical. This one construction is the generating principle of the whole system: every connective of ZTL is a lifted classical one, and every result in this preprint is a property of the lift. It is NOT the strict (Kleene-style) lift of domain theory, which propagates the extra element (f(…,⊥,…) = ⊥): the strict lift passes the mark on, the zero-trust lift interrogates it — every classical reading is taken and truth must be unanimous. Throughout the text and the companion repositories it is called simply *the lift*; its emblem is **↕** — the lift raises the operation into the marked world, the greedy collapse returns the verdict to the classical floor: the mark never lives above the ground level. (In formulas the lift stays f*; ↕ is heraldry, not notation.)

Corollary (greediness theorem, MEASURED): no compound formula ever takes the value Z; Z lives only on atoms.

Tables — both registers, all twelve (generated by the principle; the anchor cells were postulated at design time and are reproduced by the principle — `ztl.py`; every cell below is read off the kernel’s own definitions, `znot/zand/zor/zimp/zxor/zxnor` and `knot/kand/kor/kimp/kxor/kxnor`). The greedy register, which issues verdicts:

x	¬x	∧	T	F	Z	∨	T	F	Z
T	F	T	T	F	F	T	T	T	T
F	T	F	F	F	F	F	T	F	F
Z	F	Z	F	F	F	Z	T	F	F

→	T	F	Z	⊕	T	F	Z	↔	T	F
T	T	F	F	T	F	T	F	T	T	F
F	T	T	T	F	T	F	F	F	F	T
Z	T	F	F	Z	F	F	F	Z	F	F

The lazy register (strong Kleene, §9), in which the mark flows:

x	¬x	∧	T	F	Z	∨	T	F	Z
T	F	T	T	F	Z	T	T	T	T
F	T	F	F	F	F	F	T	F	Z
Z	Z	Z	Z	F	Z	Z	T	Z	Z

→	T	F	Z	⊕	T	F	Z	↔	T	F
T	T	F	Z	T	F	T	Z	T	T	F
F	T	T	T	F	T	F	Z	F	F	T
Z	T	Z	Z	Z	Z	Z	Z	Z	Z	Z

The two sets are not mirror images of each other, and neither can be read off the other by a relabelling: they part in 20 of the 48 cells, and in every one of them the greedy side answers F where the lazy side keeps Z — the greedy register never returns the mark above the atoms (the greediness theorem, §3), the lazy one returns it wherever a reading of the unverified could still change the answer, and no rule on the cells recovers one table from the other without the tables themselves (¬Z is F in one and Z in the other; F ∨ Z is F in one and Z in the other; Z → Z is F in one and Z in the other). Both are needed and both are shown, because the paper runs on both (§9): the lazy register is where grounding and the receipt live, the greedy one is where a verdict is issued.

Generating basis (MEASURED): {¬,∧,∨} generates everything: p→q = ¬p∨q, p⊕q = (p∧¬q)∨(¬p∧q), p↔q = (p∧q)∨(¬p∧¬q) are surviving identities.

Entailment: Γ ⊨ φ iff every valuation making all premises T makes the conclusion T. Tarskian by construction.

3. Results (all MEASURED)

3.1 Laws: 12 alive, 14 fallen

Alive: MP, non-contradiction, transitivity of →, commutativity and associativity of ∧/∨, both distributivities, and the three canonical definitions of the derived connectives. Fallen: ¬¬p=p, both De Morgan laws, contraposition-as-identity, ⊕=¬(↔), idempotence of ∧/∨, absorption, the units (p∧T=p, p∨F=p), excluded middle, p→p, Peirce's law, q→(p→q). The common trait of the fallen: each fails only on Z, and each is a law of "free truth" (truth from form or from polarity flip).

The three laws of thought (MEASURED). Of the classical triad, exactly one survives the lift. Non-contradiction ¬(p∧¬p) is T under every marking and hereditary (§19); identity p→p and excluded middle p∨¬p both fall on Z. One reading covers the split: **a denial is free, an affirmation is on credit.** Non-contradiction denies — "not both" is forced (p∧¬p is F under every reading of a mark) without knowing p; identity and excluded middle affirm — "p implies p", "p or ¬p" — and the lift will not affirm truth over an unverified p. The classical duality breaks along the very line the principle draws: non-contradiction and excluded middle are De Morgan twins, yet the lift keeps the one that withholds and drops the one that grants. The generating principle read on the oldest laws — truth is refused on credit, never granted on it.

3.2 The split between rules and laws

Of 14 classical inference rules taken as entailments, 12 survive — including contraposition-as-a-rule (p→q ⊨ ¬q→¬p) and the K-rule (q ⊨ p→q), whose law twins fell. Two fall: ¬¬-elimination (¬¬p ⊨ p; Z leaks through double negation) and "tautology in the conclusion" (p ⊨ q∨¬q; a fresh atom earns no truth). Classical logic cannot see the split because the deduction theorem glues it shut; in ZTL the deduction theorem works **left-to-right only**: ⊨A→B implies A⊨B, but p⊨p while ⊭p→p. The arrow is stricter than entailment: entailment transports earned truth, the arrow is a verdict that must earn its own. (The one-way-ness is a property of the primitive arrow, not of the language: a definable external implication satisfies the full deduction theorem — §3.6.)

3.3 Classification: paracompleteness

ZTL is paracomplete (LEM fell) but not paraconsistent in the ⊨ sense: v(p)=v(¬p)=T is unsatisfiable, there are no gluts, explosion is vacuously valid. Semantic MP is intact — ZTL is stronger in inference than Priest's LP.

3.4 Paradoxes: quarantine, not tables

Negation has no fixed point (¬Z=F) — the liar cannot be seated by the tables (MEASURED: enumeration of all values). Paradoxes are extinguished by the quarantine flag: Z-sentences are exempted from the Tarski schema. Quarantine is detectable from inside: isZ(x) = ¬(x↔x). The same formula expresses the revenge liar; its content evaluates to T while the sentence is denied truth — a price paid deliberately (it is the standard price of all quarantine theories; here it is written out explicitly).

3.5 Incompatibilities (mini-theorems — Lean, `Frame.lean`)

Three statements about the frame itself. Until v1.2 they were argued in a parenthesis ("verified by substitution"); they are now kernel-checked on the empty axiom list, since a paper about the difference between earned and borrowed truth should not carry the word *theorem* on unformalised reasoning about its own core.

1. **{¬Z=F, T→Z=F} ⇒ contraposition-as-identity is impossible.** At p = T, q = Z the left side is T→Z = F while the right side is ¬Z→¬T = F→F = T (`mt1_contraposition_impossible`). The dependence is stated separately as an implication from the two named cells (`mt1_from_the_cells`): the failure is a consequence of those entries, so a rescue costs exactly the flip of one of them.
2. **The quarantine flag is irremovable.** Housing the liar means being a fixed point of ¬; the frame has none (`mt2_housing_means_fixed_point`), and at Z it is pessimism, ¬Z = F ≠ Z, that does the excluding (`mt2_pessimism_excludes_Z`). Quarantine is therefore not a fourth value one could reach by editing a cell — it is a property of the input mark, not a verdict the tables could return (`mt2_quarantine_irremovable`).
3. **Collapsing Z→F at the atom is the classical fork.** The claim proved is the strong one, not "some laws come back": under the atom-collapse the ZTL evaluation of *every* formula equals the embedded Boolean evaluation, term by term (`mt3_collapse_is_classical`), whence every classical tautology returns (`mt3_every_tautology_returns`) — p→p and excluded middle among them. With the price list empty the rules/laws split has nothing left to split, and the system is Bochvar's isomorph B3☐. The fork is one step wide: the same p→p is F in ZTL proper, where the mark reaches the operator and is read there (`mt3_the_fork`).

3.6 The algebraic passport: completeness as a logic (MEASURED + Lean)

The fallen laws pay for the algebra. The chain, each link verified by total enumeration and kernel-checked in Lean (module `ZAlgebra`, empty axiom list):

1. **Fallen idempotence is a truth detector.** J_T(p) = p∧p takes T exactly at p=T (the decorrelated readings of Z kill the Z-diagonal); J_F(p) = ¬p∧¬p detects F; J_Z = isZ detects Z. Three exact disjoint indicators — the Rosser–Turquette J-operators, grown from ZTL's own connectives.
2. **Expressive completeness of the external layer.** Every external function $V^n \to \{T,F\}$ is a disjunction of indicator conjunctions over the cells of its table (MEASURED totally: 8 of 8 unary, 512 of 512 binary; the unary construction is a Lean theorem). The basis {¬,∧,∨} generates the entire external clone.
3. **The full deduction theorem returns — one floor up.** The definable external implication E(p,q) = ¬(p∧p) ∨ (q∧q) satisfies Γ,A ⊨ B ⟺ Γ ⊨ E(A,B) in both directions (MEASURED: 0 divergences on 324 triples where the primitive arrow diverges 20 times; Lean: `ddt_E` over the whole language, with premise lists). The logic does internalize its own consequence — with the meta-level "if A is true then B is true" made internal, not with the zero-trust arrow, whose one-way-ness (§3.2) stands.
4. **Algebraizability.** The same-value detector Δ(p,q) = (J_T p ∧ J_T q) ∨ (J_F p ∧ J_F q) ∨ (J_Z p ∧ J_Z q) and the truth equation p∧p ≈ ¬(p∧¬p) (which holds exactly at p=T) witness the Blok–Pigozzi conditions: ⊨Δ(p,p); p, Δ(p,q) ⊨ q; congruence for all six connectives; p ⊣⊨ Δ(p∧p, ¬(p∧¬p)). All four verified on the matrix (MEASURED + Lean); by the Blok–Pigozzi characterization [31] **ZTL is algebraizable** — its equivalent algebraic semantics is the quasivariety generated by the three-element algebra. This is family kinship, not discovery: Bochvar's external logic is algebraizable via the quasivariety of Bochvar algebras (Bonzio–Pra Baldi [32]); ZTL grows the witnesses from its own primitives, and we verify the conditions directly rather than inherit them.
5. **Yet not self-extensional.** p ⊣⊨ p∧p (idempotence is dead as a law, alive as interderivability) while ¬(p∧p) ⊭ ¬p (at p=Z): interderivability is not a congruence. The same failure that breaks self-extensionality builds the detectors — one design decision seen from two sides.
6. **Structurality.** The substitution lemma (Lean: `evalF_subst`, by induction on the formula) gives: Γ ⊨ φ implies σΓ ⊨ σφ for every uniform substitution σ. Together with reflexivity, monotonicity and cut (immediate from the definition), ⊨ is a **structural Tarskian consequence relation** — finitary and decidable, since the matrix is finite.
7. **Craig interpolation** holds, and its proof is one line on top of item 2: if A ⊨ B, the projection of A onto the shared atoms ("some completion of A's private atoms makes A true") is an external function of the shared atoms, hence a formula — and its J-DNF interpolates: A ⊨ I (A's own valuation is the witness), I ⊨ B (glue the A-witness to the current valuation; B does not read A's private atoms). MEASURED totally: 400 of 400 entailing pairs on the one-shared-atom pool, 32 of 32 on a two-atom cross-sample. The standard caveat: with an empty shared set the interpolant is a constant, and in the constant-free language ⊤ = ¬(x∧¬x) spends a spare variable — exactly as classically.

3.7 The quasivariety, scouted (MEASURED)

A reconnaissance of the equivalent algebraic semantics — the quasivariety generated by A = ({T,F,Z}; the six connectives) — before committing to its structure theory:

- **Subalgebras:** exactly two — the Boolean core {T,F} and A itself. C-extension, algebraically: the classical world is the unique proper subalgebra.
- **Congruences:** the measurement corrected our guess — A is NOT simple. The partition {T,F}|{Z} is a congruence, and it is exactly the *greediness kernel*: its compatibility IS the statement that no operation ever outputs Z. The congruence lattice is the three-chain $\Delta < \theta\_Z < \nabla$, so **A is subdirectly irreducible with monolith θ_Z** — greediness is not merely a theorem about the algebra, it is its monolith. The quotient A/θ_Z is the two-element algebra in which every operation lands in the classical class: the mark evaporates, algebraically.
- **The clone theorem (exact):** the term operations of A are *precisely* the projections plus all external functions (unary 1 + 8; binary 2 + 512, by closure computation) — §3.6's expressive completeness sharpened to a clone identity: nothing else sneaks in.
- **The Plonka probe (the structural discriminator):** Bochvar algebras [32] are built over Plonka sums, and Plonka sums of Boolean algebras satisfy every *regular* Boolean identity. A fails the regular identities $p \wedge p = p$, $p \vee p = p$, $\neg\neg p = p$ — so **A is not a Plonka sum of Boolean algebras**: the greedy quasivariety is structurally foreign to the weak-Kleene/BCA landscape. Kinship in expressive power, divorce in structure.

What a dedicated paper would need: a quasi-equational axiomatization of Q(A), its subquasivariety lattice, and a representation theorem replacing Plonka sums. The reconnaissance says the ore is there; the mining is left as a separate work.

3.8 Single-operator completeness: the census of sixteen (MEASURED + Lean)

Zhegalkin's classical theorem splits in two under the lift, and the halves part ways. The *basis* survives entirely: the clone generated by $\{\wedge, \oplus\}$ + constants equals the clone of the canonical basis $\{\neg, \wedge, \vee\}$ + constants — a kernel clone equality, with negation verbatim as $x \oplus \top$ and the disjunction witness $J\_T(x) \oplus y \oplus (x \wedge y)$, where the truth detector $J\_T(x) = x \wedge x$ (the fallen idempotence law, §3.6) repairs the classical polynomial. The *ring* falls: the unit law $x \wedge \top = x$, idempotence and the distributivity of $\wedge$ over $\oplus$ all die on Z (only the characteristic-2 law $x \oplus x = \bot$ survives — it is an anchor cell), so no multilinear canonical form exists; the normal-form role belongs to the J-DNF of the external layer (§3.6).

The census. Lifting each of the sixteen binary Boolean kernels and closing over projections and constants gives, totally:

fate	kernels	clone size
complete alone	$\rightarrow, \leftarrow, \nrightarrow, \nleftarrow$	514
fallen	NAND, NOR	18 (one shared cage)
fallen	$\wedge, \vee$	7
fallen	$\oplus, \leftrightarrow$	258
degenerate	$p, q, \neg p, \neg q, \top, \bot$	4–8

The census law: a lifted binary connective is complete alone (with constants) iff its kernel is essentially binary and non-commutative. The four directional connectives survive — both implications (the classical arrow-and-falsum basis of Hilbert systems crosses the lift intact) and both abjunctions (nonimplication $p \wedge \neg q$ and its converse $\neg p \wedge q$, the *credit detector*: its single truth cell is "the consequent stands, the ground does not" — the very situation the logic is built to forbid). Every commutative kernel falls, including both classical one-operator champions: Sheffer's stroke and Peirce's arrow stall in the *same* eighteen-table cage, each still defining negation on its diagonal ($x \uparrow x = \neg x = x \downarrow x$) yet unable to rebuild its own De Morgan partner. Directionality, not any "mark-killing", is the mechanism: the lifted implication mints forced truth over marked cells ($Z \rightarrow T = T$) and is complete regardless. Classical completeness theory is untouched — these are statements about the lifted operations relative to the lifted clone (for the classical landscape see Post; Martin [33]; Rosenberg [34]). The completeness landscape of the lifted clone itself, however, is *not* new: this clone is the external-Bochvar class B^3_{ex}, whose maximal (pre-complete) classes were determined by Finn [35, Remark 1] — the five lifted Post classes (preserving falsehood, preserving truth, self-dual, linear, monotone) together with **two additional maximal classes generated by the NOR- and NAND-type connectives**. A single binary kernel is complete alone in B^3_{ex} iff its lift lies in none of these seven; both classical champions fall precisely because each generates one of the two extra classes — the fact our census re-derives and machine-checks. Our contribution here is therefore not the completeness theory (Finn's) but (i) the explicit kernel-verified census of all sixteen binary kernels with clone cardinalities below, (ii) the observation that solo-completeness tracks non-commutative directionality, and (iii) the reading of the surviving basis ↛ as the credit detector. (Our census further finds the two champions sharing a *single* eighteen-table clone, which reconciles with Finn's two extra classes exactly. There are precisely two external negations — the mark-intolerant J_0 (the mark reads as false) and the mark-tolerant one (the mark reads as true); Finn's two extra maximal classes are their respective ∩-clones (measured, `finn_reconcile.py`). Our greedy NOR is $J_0x_1 \cap J_0x_2$ cell-for-cell, and both zero-trust champions generate its class $B^3_{ex},\neg$; the mark-tolerant negation generates the distinct $B^3_{ex},-$. Zero-trust is therefore, at the clone level, exactly the choice of the mark-intolerant negation — the unverified reads as not-true, never as true — and $B^3_{ex},-$ is the shadow it rejects. Matching Finn's $\bar{x}$ to the mark-tolerant negation is forced by elimination; its verbatim definition is in [35].)

Machine verification (`lean/ZClone.lean`, empty axiom list throughout): the negative half by the certificate method — the explicit 18-table cage, its closure under both champions a single `decide`, with Boolean membership by own recursion (the core decidability of list membership is propext-tainted); the positive half by witness terms spliced through table extensionality as data (nine cell equalities instead of `funext`), giving the kernel clone equalities clone(↛) = clone({¬,∧,∨}) = clone({∧,⊕}), plus the seven-table cage banning lone ∧ from negation. The only measured (not kernel) remainder is the cardinality of the common clone: 514 = 2 projections + all 512 external binary tables, by exhaustive closure.

4. Place in the literature

The pedigree, with the ledger kept honestly. Nothing here was taken from Bochvar constructively: the tables were generated by the zero-trust principle (§2; the anchor cells were design axioms), and the kinship below emerged only in the subsequent literature search — the repository's commit history is the lab notebook of that order. What the search found: on {¬,∧,∨} ZTL coincides verbatim with the external layer of Bochvar's B3 (1938; built against the paradoxes) — ¬ = ˥ (the ◇-negation), ∧ = ∩□, ∨ = ∪□ — and the truth detector J_T = p∧p of §3.6, derived here from the fallen idempotence, carries the table of Bochvar's *primitive* assertion operator: what he postulated, the principle re-derives. There the coincidence ends. The delta is 7 cells in →, ↔, ⊕: a systematic closing of the places where Bochvar's quarantine begets vacuous truth (his ½⊃0 = 1, our Z→F = F). The internal dynamics are opposite — his meaninglessness is infectious, our mark evaporates at the first operator — and so is the ontology. That parting need not be argued at length, because it has a name. **Suszko's Thesis** (Suszko 1977 [37]) holds that every structural logic is *logically* two-valued: the many values of a matrix are *algebraic*, administrative, while the *logical* values are only two, designated and undesignated. On that thesis Łukasiewicz, K3 and Bochvar are already two-valued as logics, their third value an algebraic bookkeeping symbol recovered as bivalent by the Suszko reduction *after the fact*. ZTL's difference is not that it escapes the reduction — nothing does — but that it needs none: it is bivalent **by construction**. The discriminator is exact and machine-measurable: does the third symbol ever appear as the value of a *compound* assertion? In the neighbours it does — 602 to 784 of the 784 depth-2 compounds over two atoms take the middle value (MEASURED). In ZTL it never does — **0 of 784**, the greediness theorem (`evalF_classical`, empty axiom list): the mark evaporates at the first operator, so no *compound* assertion is ever anything but T or F. Where a genuinely three-valued logic is shown bivalent by Suszko's reduction, ZTL is bivalent before any reduction, because the mark is barred from the value of compounds from the start. That is the precise sense in which ZTL is a two-valued logic with a mark, not a three-valued logic. ZTL's implication is Bochvar's ◇A⊃□B taken as a primitive; a polarity-adaptive translation (□ in positive positions, ◇ in negative ones) instead of a uniform one. ZTL coincides with none of the literal paralogics of the Karpenko–Tomova lattice (2017), and its position against Tomova's class of **natural implications** [38] deserves stating precisely, because the loose version — "it fails p→p" — invites a charge it does not deserve. The class is defined by four criteria: (1) the restriction to {0,1} is the classical implication; (2) normality in the sense of Łukasiewicz–Tarski [39, p. 134] — modus ponens preserves the designated value; (3) if $p \le q$ then p→q is designated; (4) anything elsewhere. ZTL satisfies (1) — proved for the whole language, not sampled, in `lean/ClassicalAgreement.lean` (`evalF_agrees`, empty axiom list) — and satisfies (2), since T→Z = F blocks the only counterexample. It violates (3), and MEASURED, in exactly **one cell**: (Z, Z). The law p→p is not a primitive requirement anywhere in the definition; it is the diagonal case of (3), and (3) presupposes a linear order on the values, i.e. that the middle symbol is a *degree of truth* standing between falsity and truth. Under that reading — Łukasiewicz's ½, "possible, not yet determined" — the condition is compelling: an antecedent no truer than its consequent should not falsify the conditional. Under ours it has no subject. Z is not a degree but a mark of status, barred from the value of compounds (the greediness theorem above); Z ≤ Z does not say "equally true on both sides", it says "neither side has been examined", and to designate the conditional there is precisely to grant truth on credit. ZTL is therefore outside the class not by breaking a law of logic but by falling outside the family the classification is built for — and the constitutive test of logicality is met elsewhere and independently: the consequence relation is Tarskian (reflexive, monotone, closed under cut) by the shape of its definition, so $p \vDash p$ holds even where $\vDash p \to p$ fails. The price of that gap is named in §3.2: the deduction theorem holds left to right only. The *meaning* of the mark, as against its tables, has a separate and earlier ancestor: Łukasiewicz's $Ł_3$ [36] (1920), the first three-valued logic, whose middle value read *possible / not yet determined* — the nearest kin to Z's *unverified until verification.* His motive was future contingents, not the paradoxes, and his implication keeps $p \to p$, so the kinship is of intent, not of tables. The two axes separate cleanly: the tables are Bochvar's external B3, the mark's meaning is Łukasiewicz's, and no system in the surveyed literature occupied their conjunction.

The *consequence* relation is separated from each of its four involutive-negation neighbours by a single rule. Against each of them (K3, LP, weak Kleene, Łukasiewicz $Ł_3$) ZTL is incomparable as a consequence relation, and the witness on the "the neighbour derives it, ZTL does not" side is the *same* for all four: $\neg\neg\mathbf{p} \vDash \mathbf{p}$, double- negation elimination as a rule (MEASURED, depth-2 pool; the witness is of size 3 and exhibited, so underivability is settled, not merely unfound). Every neighbour keeps it; ZTL breaks it, and the reason is proved rather than tabulated: an involutive negation *forces* the rule (`involution_gives_dne`, for an arbitrary negation and designated set), and ZTL's greedy negation is not involutive — $\neg\neg Z = T \neq Z$, one cell — so it breaks the rule with the involution (`lean/Signature.lean`, empty axiom list). The broken involution is thus the feature separating ZTL from each of these involutive-negation neighbours — and, by `involution_gives_dne`, from *any* three-valued matrix whose negation is involutive, an infinite class of which the four are instances. Its non-involutive kin is the exception that keeps the claim honest: external Bochvar shares the $\{\neg,\wedge,\vee\}$ tables, hence the broken involution itself, so $\neg\neg p \vDash p$ does *not* separate the two — there they part in the implication fragment (the seven cells above), not on this rule. The same greedy $\neg\neg Z = T$ (i) makes the compounds classical above, (ii) bars ZTL from *expressing* any mark-carrying neighbour — greediness lets no compound output the mark, while every neighbour's connectives do (MEASURED) — and (iii) witnesses the separation here. One feature, three faces; the novelty is that feature and the principle that generates it, not the shared external tables. The passport's two non-classical letters carry two distinct pedigrees accordingly — Z from Łukasiewicz's indeterminate (a truth not yet settled), N from Kleene's undefined (a computation not yet run, §10). Kindred in spirit: supervaluationism (rigid, non-tabular), subvaluationism/Jaśkowski (its dual), SQL NULL (lazy), IEEE NaN (comparisons), exceptions with try/catch at every node (greediness). The motivation "greedy local supervaluation / default deny" was not found in the surveyed literature. The passport spells four letters, N, Z, F, T, in the genetic order — nothing (pending), the unsettled (a question with no answer yet), the free denial (default-deny), the earned affirmation (truth on a ground): nothing → doubt → default-no → grounded-yes. Z is born second, as raw doubt, and returns last, as the hardened liar (§10) — the pre-computer genesis, not the solver's lifecycle.

Devyatkin (2016, "Non-classical modifications of many-valued matrices"). The $\{\neg,\wedge,\vee\}$ core of ZTL fits the templates of his class 8Kb* — the paracomplete duals of the 8Kb family (Carnielli–Marcos, 8192 matrices): $\neg Z{=}F$, $Z\wedge x{=}F$, $Z\vee T{=}T$ land in the permitted "0 or 1" cells. So our triple is a member of a catalogued class — a stronger pedigree than previously known. However, the entire catalogue is built under the Rosser–Turquette standardness conditions for implication: middle→0 must be designated. ZTL with $Z \to F = F$ and $Z \to Z = F$ violates these conditions deliberately — the implication/equivalence floor of ZTL lies outside the catalogue, and it is precisely this floor that carries all the system's signatures. The significance logics of Goddard–Routley (checked at survey level) use infectious nonsense plus classical external operators — Bochvar's architecture, not per-operator collapse.

Database-theory neighbours. The closest kin is Libkin's line: (a) certain answers ("true in every completion of an incomplete database") = global supervaluation, the high-complexity theoretical gold standard (Libkin, ICDT'15/TODS'16); (b) Libkin–Peterfreund, "SQL Nulls and Two-Valued Logic" (PODS'23): SQL without the third value — atomic comparisons with NULL yield false (in both polarities, the same NaN asymmetry), while the Boolean layer is fully classical, restoring all laws (their goal is optimization). On the "where does indeterminacy collapse" scale, three of four positions are taken:

where the indeterminacy collapses	system
at the atom (predicate)	Libkin–Peterfreund 2VL ≈ Bochvar's B3□
at every operator	**ZTL — position not found occupied**
over the whole formula at once	certain answers / supervaluation
never (it flows)	Kleene / SQL 3VL inside expressions

The per-operator position is the only one that yields tabularity, two-valued verdicts and the internal signatures (¬¬Z=T, Z↔Z=F) at once; the price is the rewriting laws, which the atomic position preserves (which is why the optimizers chose it). A terminological caution: “local/global validity” in the philosophy of supervaluationism (McGee–McLaughlin, Varzi) is a distinction at the level of inferences, not of operators; our “locality” is a different notion.

5. The calculus: signed tableaux (MEASURED + Lean)

Hilbert style is closed off (axiom K fell, valid formulas are scarce), the deduction theorem is one-directional — the calculus is built as signed tableaux (the Rousseau–Hähnle architecture for finitely-valued logics).

Signs — four sets of values:

- strict: **T** = {T}, **F** = {F};
- weak: **P** = {T,Z} (“possibly T”), **N** = {F,Z} (“did not earn T”).

Rules (branches separated by “|”, a comma puts both nodes on one branch):

```
T:¬φ    →  F:φ                          F:¬φ    →  P:φ
T:(φ∧ψ) →  T:φ, T:ψ                     F:(φ∧ψ) →  N:φ | N:ψ
T:(φ∨ψ) →  T:φ | T:ψ                    F:(φ∨ψ) →  N:φ, N:ψ
T:(φ→ψ) →  F:φ | T:ψ                    F:(φ→ψ) →  P:φ, N:ψ
T:(φ⊕ψ) →  T:φ,F:ψ | F:φ,T:ψ            F:(φ⊕ψ) →  P:φ,P:ψ | N:φ,N:ψ
T:(φ↔ψ) →  T:φ,T:ψ | F:φ,F:ψ            F:(φ↔ψ) →  P:φ,N:ψ | N:φ,P:ψ
```

On compound formulas P≡T and N≡F (the greediness theorem: a compound is never Z). **Branch closure:** the intersection of some formula’s signs is empty (T against N, F against P, T against F); a pair P and N on an atom does *not* close the branch — their intersection {Z} yields a Z-countermodel. **Procedure:** $\Gamma \vdash \varphi \iff$ the tableau from $\{T{:}\gamma \mid \gamma\in\Gamma\} \cup \{N{:}\varphi\}$ closes entirely.

The zero-trust signature inside the calculus. T-polarity rules demand strict certificates (only T/F); weak signs appear exclusively in F-polarity. Classical tableaux are these same rules with P≡T, N≡F glued — the whole contribution of Z is the unglueing of the negative signs. Proving truth in ZTL costs the same as classically; refuting is cheaper, because a countermodel may hide in Z.

Soundness and completeness (MEASURED): (a) each rule is machine-checked against the preimage of its table — the branches cover the preimage exactly; (b) the tableau decisions coincide with semantic ⊨ on 2462 entailments (the rule battery + all pairs of a generated formula pool + a sample of two-premise sequents). For finitely-valued tableaux with exact preimage coverage, soundness/completeness is the standard constructive result; the machine cross-check is an independent control of the implementation (and §8 upgrades it to a kernel-checked proof).

The sequent reading and semantic cut elimination. Read bottom-up, the tableau engine is a cut-free sequent calculus for the refutability judgment ⊢ S (“the signed set S is jointly unsatisfiable”): axioms are the contradictory atom constraints, the rules are the twelve signed expansions with premises above the line, derivability is the engine’s closure. The cut rule on a covering sign pair (every value lies in {T} ∪ {F,Z}):

$$\frac{\vdash S,\ T{:}\varphi \qquad \vdash S,\ N{:}\varphi}{\vdash S}\ (\mathrm{cut})$$

is sound, and the cut-free system is already complete — hence **cut is admissible**: the classic semantic cut elimination, here with a machine certificate. Kernel-checked with zero axioms (Lean: `cut_admissible`, `weakening_admissible`, `identity_refutable` on top of `closes_iff`); MEASURED directly as well (identity 14/14; weakening 696 checks, 0 violations; cut 406 fired instances on each covering pair, 0 violations). **And the syntactic procedure exists, with its bound as a function** (E55 — E-numbers name the expeditions, the numbered entries of the repository’s log — `ZCutElim.lean`, empty axiom list): derivations are an inductive calculus over the engine’s own rule table — a rule acts on ANY node, which makes exchange free and weakening size-preserving; the engine is the special case that always picks the head (`der_of_closes`). The index is SIZE, the number of closed branches: in a consuming calculus height cannot blow up, width is what a cut buys. `cutP`: a derivation of `S, T:φ` with k_1 leaves and one of `S, N:φ` with k_2 leaves yield one of `S` with at most `B0 true` `k₁ k₂ (basis φ)` leaves, where `B0` is the recursion of the procedure — an atom MULTIPLIES (`cut_env`: a cell split two ways, the two derivations merged leaf against leaf), ¬ flips the pair `T/N ↔ F/P` at no cost, ∧ and ∨ cut the two subformulas in turn with the second cut fed by the first — so a chain of `d` connectives multiplies by a leaf count per level (`bound_or_chain`, `bound_and_chain`: `2·2 → 32` at depth 4). Every step is an inversion (`inv1`, `inv2`, `inv_atom` — height-preserving, at any position), a weakening, a permutation or a smaller cut; nothing goes through a model. The bound has the classical shape: the greedy collapse changes which ATOMIC sequents are axioms, not the branching skeleton. Where ZTL shows is the PAIR: the classical cut, `T` against `F`, is refused on atoms (`tf_cut_fails_on_atoms`: `{N:p, P:p}` is open with `p = Z`, yet `T:p` and `F:p` both close it) and granted on compounds (`tf_cut_compound`), because on a compound `F` and `N` are the same two bits (`der_swap`) — the greediness theorem seen from inside the calculus. MEASURED (`zcutelim.py`: the same procedure as a program, every tree it builds re-verified against the rule table): on the 406 cut instances of the E16 pool and on 300 random instances to depth 3 over three atoms, zero bound violations, the bound met with equality in 84 and 76 cases — and the procedure's output NEVER exceeded the engine's direct cut-free derivation (equal in all 300 deep instances), while the proved bound reached 878 800 for an output of 27 leaves. So the bound is loose by five orders of magnitude on the instances that ran; 253 further instances whose bound exceeded 10^6 were not run, and the pool holds none of the hard tautologies where cut-free tableaux are known to be exponentially larger, so no improvement of the bound in general is claimed — only that on these pools the classical recursion overshoots what the procedure does.

6. Quantifiers: finite domains and beyond (MEASURED + Lean)

By the generating principle: **∀xφ = T if every instance is strictly T, else F** (one Z-witness poisons the universal); **∃xφ = T if some instance is strictly T, else F** (a Z-candidate does not count as a witness). Greediness extends: quantified formulas never take Z.

Measured over all interpretations (unary P,Q — domains 1..3; binary R — domains 1..2):

- **Identities:** both distributions survive (∀ over ∧, ∃ over ∨); both quantifier De Morgan laws fell (counterexample: a one-element domain with P=Z — Z hides under negation).
- **Instantiation asymmetry:** UI survives even as a **law** (⊨ ∀yP(y)→P(a) — the universal has earned its truth, spend it freely), EG as a law fell (⊭ P(a)→∃yP(y) at P(a)=Z); only the EG rule survives.
- **The rules/laws split continues on the quantifier floor, and the first rule casualty appears:** ¬∃yP ⊭ ∀y¬P — from "no strict witness" it does not follow that "all are strictly false": Z-elements remain. This is the quantifier twin of fallen ¬¬-elimination. The converse rule ∀y¬P ⊨ ¬∃yP survives, as does the quantifier swap ∃x∀yR ⊨ ∀y∃xR.
- **Classical ornaments:** quantified LEM fell; the "drinker paradox" ∃y(P(y)→∀zP(z)) — a classical validity — fell in ZTL (in a bar with one Z-patron there is no drinker).

Quantifier tableaux (MEASURED + Lean): the finite-domain rules continue the sign signature — T:∀ unfolds into strict T:φ(a_i) on one branch, F:∀ into weak N:φ(a_i) across branches; ∃ mirrors. Tableau decisions coincide with semantic enumeration on 28 sequents (domains 1–2). A by-product: ¬∀yP ⊭ ∃y¬P even as a rule — the second fallen quantifier bridge, symmetric to ¬∃ ⊭ ∀¬ (negation hides Z in both directions). The finite-domain quantifier tableaux are now **kernel-checked** (Lean module `ZQuant`, zero axioms): over a finite domain the quantifiers are strict folds expressible in the certified language (∀ as a conj-fold, ∃ as a disj-fold — on a singleton domain both collapse to the J_T guard φ∧φ of §3.6), the n-ary signed rules are theorems about the folds (`cover_allF_T/F`, `cover_exF_T/F`), UI/EG hold in membership form over the whole language (`ui_mem`, `eg_mem`), and eight battery verdicts — including the failing drinker and quantified LEM — are kernel evaluations of the certified engine itself.

Arbitrary domains: parameter tableaux (MEASURED + Lean). Over arbitrary domains the finite unfolding is unavailable; the standard cure — parameter (free-variable) tableaux with γ/δ rules — carries the zero-trust sign discipline over exactly:

```
γ (reusable, every parameter):  T:∀xφ → T:φ(c)     F:∃xφ → N:φ(c)
δ (fresh parameter, once):      F:∀xφ → N:φ(c*)    T:∃xφ → T:φ(c*)
```

Fresh witnesses appear exactly where the propositional calculus allows weak signs (F-polarity) or demands a strict witness (T:∃). Status, honestly split: **soundness is proved for the rules and the search in the Lean port (§27; the classical rule F:∀ only outside the corpus) and measured for this engine** — every sequent the engine proves is re-checked by total enumeration over finite domains, every saturated open branch yields a countermodel that is verified by evaluation (battery of 13: UI/EG, distribution, quantifier bridges, swap and its failing converse, the failing unguarded drinker and quantified LEM — all 13 verdicts confirmed).

And the axiom tier of the four rules themselves is now measured, before the port rather than during it (`ZParamSound.lean` for three of them, `inventory/ПАРАМЕТР-ЯРУС.py` for the fourth). Both γ rules and the δ rule for `T:∃` are sound on the EMPTY axiom list: a universal delivers every instance, `¬∃ → ∀¬` is constructive, and eliminating an existential to interpret a fresh parameter chooses nothing. The fourth, `F:∀xφ → N:φ(c*)`, is not: to interpret its fresh parameter one needs an instance that is not strictly T, and the premise gives only that not every instance is — which is `¬∀ → ∃¬`. Proved classically it carries `propext, Classical.choice, Quot.sound`. *Per-point decidability does not rescue it:* `V` has decidable equality, so each `ψ(d) = T` is decidable, but the quantifier over an arbitrary domain is not — the obstruction is the survey of the domain, which is what this logic declines to call an act everywhere else. *And the split falls where §6 already said the logic breaks:* `¬∀yP ⊭ ∃y¬P` is listed above as the second fallen quantifier bridge, so the calculus's own metatheory needs, at exactly one rule, the step the object logic refuses. The classical proof is kept OUT of the corpus rather than exempted in the audit — the corpus carries one invariant this paper leads with, and an exemption added to silence one alarm is where the next one hides; it lives in `inventory/probes/` with a stand that fails in BOTH directions, red if that rule ever becomes constructive (a finding, not a nuisance) and red if any of the three stops being clean. What is NOT claimed is that no choice-free route exists: it is not found by the standard argument, and that is all the measurement says.

Completeness splits in two, and its finite half is now a theorem on the empty axiom list (E54, `ZParamHintikka.lean`). The standard Hintikka-saturation argument for finitely-valued signed tableaux [27] has two halves: a run that STOPS OPEN yields a countermodel read off the open branch, and a run that NEVER STOPS yields one on an infinite domain. The first is proved: `stuck_refutes` — for closed Γ and φ, a `stuck` run of the engine with δ_2 exhibits a total model over `Nat` making every premise T and φ not T. The model is finite (its values are the parameters on the branch, every natural normalised onto them, so the universal quantifier is a fold and totality is COMPUTED), and it is not classical: an atom carrying only a weak sign, or none, takes the mark Z — the fallen bridge `¬∀xP ⊢ ∃x¬P` is refuted with `P(c*) = Z`. Each rule is read backwards through the tables, and the cell the proof stands on is `Z ∨ Z = F`: under a Kleene lift the weak-sign rule `F:(φ∨ψ) ⇒ N:φ, N:ψ` would not be complete. And δ_2 — classical in soundness, above — is FREE in this direction: one instance under N refutes the universal by determinism (`holds_det`), with no `¬∀ → ∃¬`. One rule, classical one way and constructive the other; that asymmetry is the measurement. The second half stays argued, with its parts named: a fair strategy (the engine has one since E54, §27), König's lemma, and a total model over an infinite domain — the survey ZTL declines to call an act, the same step δ_2's soundness costs. Two honest FO phenomena appear on cue: on invalid sequents whose branches spawn witnesses forever (the unguarded drinker; the converse quantifier swap) the tableau does not terminate and invalidity is certified by a finite countermodel instead; and FO-ZTL is **undecidable** — the J-guard translation $P \mapsto P \wedge P$ (§3.6) makes every atom classical, embedding classical first-order validity (the guarded drinker is ZTL-valid and needs the classic two γ-rounds). The tableaux give semi-decidability, as classically.

7. Limitations and honest caveats

- No new three-valued functions exist or are claimed (functional completeness — Finn); the contribution is the choice of primitives and the principle. Positively: the chosen primitives generate the whole external clone (§3.6), so ZTL is the external layer presented by different primitives — its abstract metatheory (algebraizability included) is a matter of verification, not invention.
- "Deviation from classical logic" is merchandise here, not defect: on classical inputs the deviation is zero (C-extension); all exotica is the price of the policy toward Z.
- The price list is not negotiable item by item: the fallen laws are consequences of three fixed design forks (¬Z=F; greedy collapse; Z↔Z=F); regaining any law requires flipping a fork (see §3.5).
- ZTL owns no physical clock: duration, tense and "how long" are outside its axis; the only clock is the arrival of ground (§21).

8. Machine verification in Lean 4

The core is ported to Lean 4 (v4.29.1, no mathlib): the connectives are generated by the lift (tables are not postulated — computed), the anchor cells are turned from postulates into theorems, the 12 alive and 14 fallen laws, semantic MP, the greediness theorem, the homelessness of the liar (∀v, ¬v ≠ v), the isZ detector and the quantifier UI/EG asymmetry — all proven.

Axiom status: the empty list, audited exhaustively. `#print axioms` over the whole corpus returns "does not depend on any axioms": no Classical.choice, no Quot.sound, not even propext; pure computation. This is the strictest possible tier.

The claim used to rest on hand-placed prints — 158 of them at the time, against the then-current corpus of 371 — with the rest covered transitively — a sound argument (a lemma carrying an axiom infects every theorem that uses it), but an argument, and one that an unused orphan theorem would escape. It is now a measurement: `inventory/axiom_audit.py` extracts every theorem name from every module, generates one `#print axioms` per name, and fails if a single line reads otherwise. **1112 of 1112 clean**, re-run by CI on every push. The same stand refuses a module that carries theorems and is built by no target — the failure mode that let one module (`QuantumWitness.lean`) go unchecked by any automation until 2026-07-20. Three modules of the corpus — `Layering`, `RelianceBridge` and `ZReconverge` — serve a downstream consumer's reliance layer (whether a proof, and a fan-in of proofs, is eligible for reliance); they are counted and audited here, and not discussed in this paper.

The temporal modules (v1.2): `ZTime.lean` — the verification tree, with absorption, arrow and ladder-inclusion proven structurally for every formula and marking; `EpochBoundary.lean` — the epoch boundary theorem and the separation witness (§§21–22). Both self-contained, both on the empty axiom list, both verifiable from zero by a bare `lean` call.

Part II (same file): the entailment rule battery (11 alive here — modus ponens is Part I's semantic MP — + 2 fallen, including the split contraposition-rule vs contraposition-law), no-gluts, the lazy Kleene register with proven monotonicity of all connectives in the information order, non-monotonicity of the greedy register, the liar's home in the lazy register (knot Z = Z — rfl), the absence of a greedy carousel model (all 9 pairs), lazy grounding, and the computed revenge bullet.

Part III — the tableau pillars over the whole language. An inductive formula type Fm with evaluation; **pillar 1**: greediness is proven for the entire language (every compound formula is classical under every valuation — by constructor analysis, not by battery); **pillar 2**: the preimage coverage of each of the 12 tableau rules — as ⟺-theorems for arbitrary subformula values (`cover_*`).

The trade certificate (lean/TableauCert.lean). The tableau engine is formalized in full: working rules over the generating basis {¬,∧,∨} (weak signs only in F-polarity), atoms handled by constraint intersection, the heavy connectives →,⊕,↔ reduced to the basis by the surviving identities (imp_def/xor_def/xnor_def — theorems of the core). Proven:

- `closes_iff` — **soundness and completeness**: the tableau closes ⟺ the signed nodes are unsatisfiable (induction on weighted size, for all formulas);
- `tproves_iff` — **the entailment certificate**: $\Gamma \vdash \varphi$ by the engine ⟺ every valuation making the premises T makes the conclusion T.

Six smoke runs of the certified engine coincide with the measured results (⊬ p→p, MP ⊢, ¬(p∧¬p) ⊢, ⊬ LEM, contraposition-rule ⊢, ¬¬-elimination ⊬). **The certificate's axiom status: the empty list** — same as the core. This was achieved by disinfecting every known source: structural recursion on fuel instead of well-founded recursion (the WF machinery pulls in propext/Quot.sound), signs as functions V→Bool instead of lists (Lean core's list-membership decision procedure carries propext), a recursive satisfaction predicate instead of ∀-membership, combinator chains of Iff instead of rewriting by equivalences (rw with an Iff applies propext), and hand-rolled Nat arithmetic instead of omega (omega carries propext and Quot.sound). **The native engine (TableauCertN.lean):** the engine with the native signed rules for →,⊕,↔ is certified by the same induction (closesN_iff), and the theorem engines_agree shows both engines return identical verdicts; the former footnote about their equivalence is closed by a theorem. Additionally (all with zero axioms): the Lean port of marked sets (§12: a mark earns membership nowhere; a marked set is not provably a subset of itself; {Z,Z}≠{Z}; |{Z}|=[1,1]) and a corpus of facts — domain-2 quantifiers (∀=zand, ∃=zor: the UI law alive; EG, ¬∃⊭∀¬, ¬∀⊭∃¬, quantified LEM, the drinker — fallen) and dynamics (liar period 2, carousel period 4 with no fixed points, Curry homeless greedily and grounded lazily, cycle parity for lengths 2/3, the Yablo-3 truncation with a unique grounded model, the nullity of the crocodile's deal). **The algebraic passport (`ZAlgebra`, zero axioms):** the J-indicators, unary expressive completeness as a single theorem over all 8 target tables, the full deduction theorem for E over the whole language with premise lists (`ddt_E`), all Blok–Pigozzi witnesses (Δ-spec, reflexivity, detachment, congruence for the six connectives, the truth equation, condition (iv)), the failure of self-extensionality, the substitution lemma and structurality of $\vDash$ (`entails_structural`). **The sequent reading (`ZSequent`, zero axioms):** cut admissibility on top of the engine certificate (`cut_admissible`), admissible weakening, derivable identity — the semantic cut elimination of §5, kernel-checked. **Quantifier tableaux (`ZQuant`, zero axioms):** finite-domain quantifiers as strict folds, the n-ary signed rules as preimage-coverage theorems, UI/EG in membership form, and the battery of eight tableau verdicts as kernel evaluations of the certified engine (§6). **General Knaster–Tarski (`ZGround`, zero axioms):** the lazy register over the whole language, monotonicity, the least fixed point by bounded iteration, and the absoluteness of the grounded part (§9). **Expedition twins (`ZExped`, zero axioms):** streams — the equality atom never earns T, one finite witness earns apartness and it persists, Cantor's diagonal earns strict non-membership against every registry entry (§13); one marked pair collapses the injectivity certificate for every function including the identity (§14); interval decorrelation and unearned self-identity of a nondegenerate mark (§15); Dempster–Shafer thresholds (§16); atom verdicts as $\Box/\Diamond$ thresholds with the $\neg\neg$-cell separating the local ladder from global supervaluation (§17). The certified language now carries the constants $\top/\bot$ (the engines and both certificates extended, the sequent/quantifier/algebra modules unaffected), which closes the last expedition remainder: **Russell's grounding half is kernel-computed** — `lfp RUSSELL = [F,F,T,F,T,F,F,F,Z]` by the certified iteration of §9: eight membership facts ground, exactly R∈R stays quarantined (`russell_grounded`, `russell_verdicts`); $\vdash \top$, $\bot \vdash \varphi$ and $\nvdash \bot$ run through the certified engine. **The receipt trio (`Receipt`, `Linear`, `LabelExact`, zero axioms):** the label's completeness in both registers, the no-loss theorem at multiplicity ≤ 1, and the label's exactness on linear claims with a proved counterexample showing the hypothesis is load-bearing (§19). **The stitch (`bridge.py`):** one questionnaire, two engines — 609 kernel-computed answers (both registers cell by cell, the label battery of §19, the J-operators, E and Δ, certified-engine verdicts on a shared propositional and quantified battery with constants, lazy lfp of the zoo up to the nine-fact Russell system) compared mechanically against the Python stands on every regression run: zero divergences. Two further disinfection pitfalls surfaced here: an overlapping wildcard row in a match taints the DEFINITION itself with propext through the compiled matcher (invisible to theorem-level axiom prints — `kand`/`kor` were rewritten with explicit cells, and the corpus now prints definition-level axiom checks too), and core's `List.length_map`/`length_replicate` are simp-proved and carry propext — replaced by hand-rolled inductions (likewise the core Int order lemmas, omega-proved: general interval statements live over Nat, Int stays for computation).

9. Quarantine as a fixed point: the two-register architecture (MEASURED + Lean)

The quarantine flag of fork 2 in §3.5 is formalized à la Kripke. A system of sentences with a truth predicate (λ: ¬Tr(λ) etc.); the "jump" J re-evaluates the sentences under the current valuation; fixed points of J are self-consistent valuations. Two registers with DIFFERENT negations are put on trial: the greedy one (ZTL tables, verdicts: ¬Z = F — "not earned") and the lazy one (strong Kleene, the solver: ¬Z = Z — "do not judge the uncomputed"; Z flows through connectives). What follows shows these are not two candidates for one role but two mandatory different roles.

Enumeration results (the zoo: liar, truth-teller, Jourdain's carousel, the even cycle, a grounded chain, the avenger):

1. **The greedy jump is non-monotone** in the information order (witnesses found on every system) — the Knaster–Tarski argument does not apply to it.
2. **On odd cycles the greedy jump has no fixed points at all**; the iteration oscillates: liar — period 2 (F→T→F…), carousel — period 4 (FF→FT→TT→TF), avenger — period 2. The oscillations of revision theory (Gupta–Belnap) arise here not as a postulate but as the behavior of the greedy iteration.
3. **The lazy jump is monotone** and has a least fixed point everywhere: grounded sentences receive classical values, paradoxical ones — Z. The even cycle: three fixed points ({T,F}, {F,T}, {Z,Z}), the least being mutual quarantine (underdetermination, not paradox).
4. **Verdicts are greedy, read over the finished point**: the content of the avenger μ at grounded μ=Z evaluates greedily to T, while μ is denied truth — the "bullet" is paid inside the formal construction, not by a disclaimer.

Conclusion (the architectural result of this stage): the two-register design is a necessity, not a convenience. The greedy register cannot ground itself (no fixed points on the liar); the lazy one cannot pass verdicts ("exactly falsehood"). Quarantine := the Z-set of the lazy jump's least fixed point; ZTL := the greedy reading on top. The engineering precedent — SQL (Kleene inside expressions, forced falsehood at the WHERE boundary) — turns out to be not an analogy but the same theorem, found by practice.

The quarantine passport (MEASURED). Z alone is blind: the liar and the truth-teller land in quarantine with the same mark. The passport cures the blindness without touching the logic — verdicts, tables and greediness are intact; it is solver-side metadata computed per strongly connected component of the dependency graph. Kinds: **PARADOX** — the component has no classical model consistent with its grounded environment (odd cycles; the greedy oscillation period is recorded: liar 2, carousel 4) — the refusal is *permanent*; **INTRINSIC** — exactly one classical model exists (Kripke's intrinsic value): ungrounded, yet uniquely consistent — the stipulation is forced, not chosen; **UNDERDETERMINED** — classical models exist (≥ 2: even cycles, the truth-teller) — the refusal stands *until stipulation*; **INPUT** — a plain unverified datum — the refusal stands *until verification* (§19); **DOWNSTREAM** — inherited quarantine with the culprits listed (the provenance of refusal, §14 again). The operational content is the **stipulation theorem** (MEASURED totally on a mixed zoo carrying every kind at once, with the grounded part untouched): a component carries classical models (INTRINSIC or UNDERDETERMINED) iff stipulating any of them grounds it cleanly — the forced choice and the free one obey the same mechanics — and it is PARADOX iff every decree contradicts the component's own definitions: the liftable and the permanent, mechanically separated. The passport is thereby a *biography* of the mark — there are exactly three ways to acquire it, and liftability follows genesis: **born** with the datum (INPUT — lifted by verification), **hardened** out of a solver phase that completed without resolving (INTRINSIC — lifted by the forced stipulation; UNDERDETERMINED — lifted by a chosen one; PARADOX — lifted by nothing), or **inherited** (DOWNSTREAM — lifted with the culprits). "Completed" is a theorem, not a hope: the lazy iteration provably terminates within n+1 steps (§8, `ZGround`), so the phase N always dies — the liar is never "still computing"; the solver's verdict on it is final, and what is eternal is not the process but the refusal. (Contrast revision theory [7], where the process itself never settles.) The parity theorem of §11 re-derives through the passport (62 of 62 cycles), and Russell (§18) reads: R∈R — PARADOX, permanent; the twin S∈S — UNDERDETERMINED, awaiting an external decision; eight facts grounded. This is Kripke's taxonomy plus revision-theoretic signatures [5, 7], packaged as a computable instrument; the refusal classes now mirror §19, and quarantine = (Z, passport). The caveat that stood here — "Yablo stays invisible: every finite truncation is grounded, so the passport of infinite regress needs an infinite instrument" — is now discharged, and by a theorem rather than by a passport (`ZYablo.lean`, empty axiom list, §11). The instrument for the limit is not a procedure at all; the infinite system simply admits no verdict assignment, and that is proved in one step. What the caveat got right is that no finite stage could show it: every truncation is grounded, at every n, and that too is now a theorem rather than a check at n = 3.

The architecture, kernel-checked in general form. The two-register theorem no longer rests on per-instance measurements: the Lean module `ZGround` (zero axioms) proves, for EVERY finite system of definitions, that lazy evaluation is monotone over the whole language (`evalK_mono`), hence the lazy jump is monotone (`jumpL_mono`); the iteration from ⊥ ascends and stabilizes within n+1 steps at the least fixed point (`kt_fixed`, via an information-measure pigeonhole — no classical choice, no well-founded machinery); the least point lies below every fixed point (`kt_least`); and a coordinate grounded in the least point carries the same classical value in every fixed point (`grounded_absolute`) — quarantine is well-defined, machine-checked. Together with the greedy register's kernel-checked non-monotonicity (§8, Part II), **both halves of the necessity argument are machine-checked**: the greedy register cannot ground itself, and the lazy one is exactly what grounds. The inference joining them — therefore two registers are necessary, not merely convenient — is ours, not the kernel's: no Lean object states it. We say so because the distinction is the subject of this paper. A composition of theorems is worth what its parts are worth; it is not a further theorem.

10. The ontological status of Z: the system's passport

The final and most precise formulation of what has been built:

```
Truth values:    T, F                    (verdicts are always two-valued)
Input mark:      Z "unverified"           (a property of data, not truth)
Solver state:    N "not yet computed"     (a computation phase, present
                                           only under self-reference;
                                           provably finite — §9 — and
                                           never escapes outward)
Reading policy:  local, default deny      (the three-symbol tables are
                                           the policy's calculator)
```

ZTL is a two-valued logic that refuses to lie about the unverified. Two-valuedness of the values does not mean classicality: the entailment relation is provably different (LEM fell, the deduction theorem is one-directional, $\neg\neg p \nvDash p$ — §§3–5). A logic is defined by its entailment, not by its palette.

The passport on three axes. The comparison with classical logic runs in three different directions at once, and collapsing them into "stronger" or "weaker" loses all three.

- *Conservative extension by data.* On a mark-free valuation ZTL agrees with classical logic formula for formula (`ClassicalAgreement.evalF_agrees`) — where nothing is unverified, nothing changes.
- *Strict contraction by law.* Every ZTL validity is classically valid, and not conversely: `p → p` is a classical tautology and fails here on a marked atom (`ztl_taut_is_classical`, `not_conversely`). Strictly fewer laws.
- *Strict expansion in expressive reach.* The clone is exactly the projections plus the external functions — 1 + 8 unary, 2 + 512 binary, nothing else sneaking in (§3.6, §3.7). Those external functions speak about the **status** of a ground, which classical logic has no object to speak about at all; and single-operator completeness is *lost* in exchange (§3.8: NAND and NOR each stall; only the four directional kernels →, ←, ↛, ↚ survive, with ↛ read as the credit detector).

So the honest one-liner is neither "stronger" nor "weaker": **wider in subject, narrower in law, identical on verified data.**

Local versus global reading of the mark. "Z is a mark on the atom" admits two readings of verdicts: the global one ("a formula is assertable if true under all substitutions into the marked atoms at once" — classical supervaluation) and the local one ("every operator consults the mark on the spot" — ZTL). The anchor cells separate them unambiguously: on ¬¬Z (ZTL: T, globally: F) and on Z↔Z (ZTL: F, globally: T); the other five cells coincide. The global reading returns all classical tautologies but loses tabularity (supervaluation is not truth-functional) and both signature cells — the ladder of floors and the NaN signature "not equal to itself". The anchors choose locality.

The global reading now has a branch of its own, with theorems. Asked by a downstream consumer whether a *partial disclosure* suffices for the conclusion drawn from it, the global reading becomes an operational question: does the verdict survive every admissible completion of what was withheld? `lean/ContextClosure.lean` answers it for this calculus, on the empty axiom list. The local and global readings **coincide exactly** where the withheld atom occurs under no negation, no antecedent and no xor/xnor (`closure_coincides`), and provably part company outside it — the separating case being `¬¬Z`, the very cell named above (`outside_fragment_fails`). No condition on the formula *alone* can be exact, because the property belongs to the pair (formula, disclosure) (`no_syntactic_characterisation`). Rewriting into a normal form restores agreement in one direction — no verdict is then granted that a completion could defeat (`normal_form_sound`) — at the cost of verdicts every completion upholds, `b ∨ ¬b` being the witness (`normal_form_incomplete`). Prior art for the *problem* is fourteen years deep in attribute-based access control (attribute-hiding attacks, PTaCL, POST 2012; policy resistance certified in Isabelle, 2013; extended evaluation, 2019); what belongs to this calculus is the behaviour of the **greedy** lift. The contrast with PTaCL is not that its logic is immune — it is where the collapse is authored. PTaCL's unary operators, read from its own table (Fig. 1(e)), are `¬⊥ = ⊥` and a separate ~ with `~⊥ = 0`: turning an unverified ground into a false one is an explicit act the policy author performs locally, by writing `opt`. Under the greedy lift the same collapse is the semantics, applied everywhere and unwritten. So the exposure is not their oversight and not our discovery; it is the price of a default, and it is bounded in their calculus by where `opt` appears and unbounded in ours.

Why not four values. The temptation to include N as a fourth value (precedent: Codd's two NULLs for RM/V2, rejected by industry) is declined: values are not multiplied, non-values are typed. N is the ⊥ of the fixed-point iteration (§9), the pending of any promise: it exists only inside the solver and never returns outward. Read N as **Not-yet** — Kleene's undefined *by intent*, at last housed as a phase rather than a value. The four-valued algebra {T,F,Z,N} with the lazy lift over ZTL is coherent and factorizes into our two registers ({T,F,Z} fragment = ZTL, {T,F,N} fragment = Kleene), but it packs two phases into one type — a monolith instead of modules; it is kept as a possible appendix, not as the core.

Kleene, read through this passport. Kleene's third element was epistemic by intent — "undefined, not yet computed" — but his logic has a single register, so the status had no home except inside the value algebra, where it was forced to flow (¬N = N: negation passes the unknown on). Typed by our passport, his element conflates the two non-values: the *mark* on a datum (external, static, lifted only by the act of verification) and the *phase* of a computation (internal, dynamic, lifted by the iteration itself — and hardening, when it never resolves, into the quarantine mark with its own passport of kinds, §9). Strong Kleene logic is what results when both non-values are made to share one symbol *as a value of assertions*; SQL NULL's notorious ambiguity — "unknown", "not applicable", "pending" in one symbol — is the same conflation observed in the wild. The fault is not the shared symbol (our own solver reuses Z positionally during iteration) but the promotion of a status to a truth value: ZTL splits the *role*, not the alphabet, and revokes the status's right to be what a statement evaluates to.

A closing note on errors: the system has no error letter, and none is missing. An *error* is an interface event — the premature read of a phase: any value returned for a still-pending N is wrong whatever it is, because nothing has been computed yet. The engineering cousin pair makes the split visible: IEEE's quiet NaN behaves like the mark Z (carried, tested, lifted by verification), while the signaling NaN is the alarm on touching the phase. The alphabet stays four letters; E is hardware, not logic.

That last sentence is about the alphabet of the LOGIC, and it does not change. What it leaves open is whether the seam itself has letters, and a candidate register is recorded outside this system rather than inside it (in a private design note, staked and deliberately unimplemented): two marks — `M`, no ground was ever offered, and `O`, the world's own indeterminacy — beside two states, `E` for the world's silence and σ for the world's answer. The register reads ONE WAY, from the physical into the operational: a boundary letter arrives and becomes an internal one, so the four letters above are what the seam delivered rather than a second alphabet competing with them, and each is paired by what it is ABOUT — `E` with the phase `N` it reads too early, `M` with `F`, σ with `T`, `O` with `Z`.

Read that way, this section's own claim is explained rather than weakened. `E` carries no logical content of its own precisely because it is an interface accident concerning a phase that has no value yet — not a value of any kind, and so not a letter the system is missing. The other three arrows do carry content, `O` most sharply, since ontic vacancy (a value that does not exist) is not the epistemic mark `Z` (a value we have not looked at) — a distinction this repository measured on hardware before it had a name for it. None of the register is built, and it is mentioned here only so that a reader meeting `E` does not take it for a fifth truth value.

Consequence for positioning: ZTL's neighbours are not the many-valued logics but the two-valued assertability policies of the supervaluation family — from which it differs by locality, tabularity, and greedy collapse.

11. Expeditions: Curry, parity, Yablo, the crocodile (MEASURED + Lean)

Paradox as an operator — the expeditions are one construction (MEASURED, `pengine.py`). A self-referential net is a system $S_i = f_i(S_1 \ldots S_n)$; to read it is to ground it (§9). So a paradox is *paradox(f) = ground(S = f(S))* — feed an operator, form its self-reference, ground it — and the specimens below are that one construction on chosen operators: the liar is paradox(¬S), the truth-teller paradox(S), Curry paradox(S→⊥), Russell's propositional shadow again paradox(¬S). Three measured layers read a net, coarse to fine:

1. *Grounding* — the zero-trust verdict (§9): every pure self-reference lands in Z. Coarsest; it does not separate paradoxes.
2. *Solutions* — the classical models of the net: 0 = contradictory, ≥2 = underdetermined, 1 = determined. This is the content of the parity theorem below — an odd ring has no model ⟺ its reference graph is not 2-colourable.
3. *Dynamics* — the period spectrum of the greedy jump: finest. The period-1 points are exactly the solutions, so dynamics refines (2); the higher periods separate what the model count merges — the liar and Curry are the *same* 2-cycle, the k-ring carries 2k, the crocodile a 4-cycle.

Grounding is strictly the most conservative of the three: measured over all 9015 one-sentence nets up to six symbols, it reaches a classical value only when the net has a *unique* model that is also reachable from ignorance; 1068 of them have a unique classical model yet ground to Z (e.g. S = S∨¬S — classically ⊤, here Z: zero-trust will not grant a truth it cannot ground). The ZTL-settled nets are thus a strict subset of the classically-categorical ones.

Curry without negation. c = (Tr(c) → ⊥) is the paradox that breaks naive paraconsistent theories (it uses no ¬, so taming negation does not help). Measured: no greedy fixed points, iteration of period 2, lazy grounding gives Z. **The same mechanics as the liar**: quarantine does not care which operator a sentence used to invert itself — only the non-existence of a fixed point matters. Our construction silences Curry for free, where LP must sacrifice contraction.

The parity result — total (MEASURED). All cycles of length 1–5, all inversion patterns (62 systems): classical models exist ⟺ the number of inversions around the ring is even (the XOR-sum of the edges is 0). Odd rings are carousels (liar n=1, Jourdain n=2), even rings are truth-tellers (underdetermination).

Yablo: a third source of ungroundedness. s_i = "all s_j, j>i, are false" — a paradox without a single cycle. Measured: **every finite truncation is fully grounded** (the unique model F…FT, empty quarantine, exactly one greedy model). Yablo's paradox lives only at actual infinity — a finite instrument cannot see it in principle. So there are three distinct sources of ungroundedness: the odd cycle (the liar), the odd infinite progression (Yablo), and the underdetermination of even structures (the truth-teller); the first is caught finitely, the second only in the limit.

The crocodile. "I shall return the child ⟺ you predict what I will do"; the mother: "you will not return it". Formalization: R = Tr(M), M = ¬Tr(R) — Jourdain's carousel in disguise (cycle 2, one inversion, odd). MEASURED: no greedy models, iteration of period 4, mutual quarantine under lazy grounding, and — the key measurement — **the greedy verdict on the deal itself, R↔M, at the grounded point: F**. The zero-trust reading of the ancient dilemma: the deal's condition cannot be grounded — the contract is void, no obligation ever arose; the crocodile is not "unable to comply" but "never contracted". The control case (an optimistic mother, M = Tr(R), an even cycle): classical models exist — (T,T) and (F,F), the word is kept in both — but the least fixed point is still quarantine and the deal's verdict is still F: an enforceable self-referential contract does not self-enforce — which model realizes is decided by an external choice, not by logic. The difference between paradox and underdetermination is the difference between "the contract is void" and "the contract is valid but requires the parties' will".

Kernel-checked (`Facts.lean`, zero axioms). Every dynamic fact of this section is a theorem, not a battery entry: Curry is homeless in the greedy register without any negation (`curry_homeless`) and finds a home in the lazy one (`curry_kleene_home`); the liar oscillates with period two (`liar_period2`) and Jourdain's carousel has no fixed point and period four (`carousel_no_fp`); the Yablo truncation at n = 3 has the unique grounded model F, F, T (`yablo3_unique`); and the crocodile's deal is void at the grounded point (`crocodile_deal_void`).

Yablo at the limit (`ZYablo.lean`, empty axiom list). Three theorems replace the single truncation check. EVERY finite truncation is grounded, for every n, with exactly one model — the last sentence true and all earlier ones false — so the finite instrument is not too weak to see a paradox; there is nothing at any finite stage to see. The INFINITE system admits no verdict assignment whatever (`yablo_greedy_homeless`). And it is satisfiable in the lazy register, where everything unverified is admissible (`yablo_lazy_home`) — the diagnosis the liar and Curry receive, reached here for a system no finite stage could diagnose.

The classical step is avoided, and that is the point of doing it here. The textbook argument moves from "not every later sentence is false" to "some later one is true", which is `¬∀ → ∃¬` and would have taken the file to the classical tier. It is unnecessary: from "every j > i is false" it already follows that every j > i+1 is false, so the fixpoint condition makes `s_{i+1}` true outright, contradicting its falsity. No witness is extracted. A paradox of infinite regress, refuted without a single classical step.

Two things are assumed rather than derived, and the module says so: bivalence of the sentences is written into the admissibility condition as a clause — it is §6's greediness, but posited here rather than re-derived from an evaluation of an infinite quantifier, which would need the infinite half of completeness for the parameter tableaux, the one part of that port §27 still lists as open; and the rendering of $\mathtt{s}_{\mathtt{i}}$ as "T exactly when every later sentence is F" is the strict universal of §6 with the greedy denial inside it. The truncation theorem is the positive control: an impossibility result is worthless if its definition is unsatisfiable by construction, and the same shape restricted to any finite n has exactly one model.

12. Sets with unverified elements (MEASURED + Lean)

Sets are not postulated — they are derived: element equality is an atom (T/F for verified elements; Z whenever a mark is involved, including a mark with itself), membership is an ∃-fold, inclusion an ∀-fold, set equality mutual inclusion; everything computes through the core tables, with not a single special rule. Representation: (a core of verified elements, a quarantine multiset of marks).

Measured: {Z,Z} ≠ {Z} (merging is not earned — two unverified witnesses are not one); **Z ∉ {Z}** even for the same mark (SQL: NULL IN (NULL) is not true); on clean sets the whole classical set theory is intact (C-extension); a mark fells exactly the **identity laws** — idempotence S∪S=S and S∩S=S, self-subtraction S∖S=∅, reflexivity S=S and S⊆S — the same families that fell in the logic: sets inherited the price list from the tables.

Cardinality is an interval: |{1,2,Z}| ∈ [2,3], the exact value is not earned (verdict F); but |{Z}| ∈ [1,1] — **cardinality is earned even where identity is not**: one mark is exactly one thing. Cardinality and identity have split into different currencies of trust.

SQL's inconsistency is not inherited (a theorem now: `ZNull.two_equalities`, §27): SQL holds NULL≠NULL in comparisons yet merges NULLs in DISTINCT/GROUP BY — swapping equality of values for equality of marks inside one syntax. Here the core deduplicates classically and the marks live with multiplicity — each operation honest about its own business.

Kernel-checked (`ZSets.lean`, zero axioms). The load-bearing claims are theorems: a mark belongs to nothing, itself included (`memZ` — SQL's NULL IN (NULL)); a marked set is not a subset of itself and not equal to itself (`sub_marked_false`, `seteq_self_marked`), so the identity laws fall exactly as they do in the tables; and on clean sets membership, inclusion and equality are classical (`memL_classical`, `subL_refl_clean`, `seteq_refl_clean`) — the C-extension, proved rather than sampled. And the cardinality interval now has its general law (`ZTaint.card_earned_iff`): it collapses to a POINT precisely when there are no marks at all, or exactly one mark over an empty verified core. So $|\{Z\}| = [1,1]$ is not a curiosity but the second of exactly two cases — one mark is exactly one thing, and two marks are not two things. The four instances above are consequences of it rather than samples. The restatement used is proved equal to the corpus's own `cardLo` rather than assumed so, which is the whole point of stating it.

13. The reals: two failures of enumeration (MEASURED + Lean)

A real-in-the-making is a stream of digits; at time t a prefix is verified. Stream equality is an atom with a pinned fate: prefixes diverged — **F earned** (apartness, a finite witness); they agree — **Z**; **T is never earned at any t** (the comparison is infinite). Finitely-presented objects (fractions p/q) are the contrast: equality decides finitely, the atoms are T/F.

Measured — uncountability splits into two distinct impossibilities:

1. **Non-registrability (zero-trust, about presentation).** Membership of a stream in any registry — an ∃-fold of atoms from {F,Z} — is eternally F: even streams literally standing in the list (including a duplicate!) earn no membership. No enumeration of streams can certify coverage of a single element — including its own rows. Registries of fractions, by contrast, certify every element: the countability of $\mathbb{Q}$ is the earnability of presentation identity.
2. **Incompleteness (Cantorian, about cardinality).** The diagonal is an apartness-earning machine: against the i-th entry a finite witness is found by time i+1. The diagonal's non-membership is not postulated — it is earned.

Classical usage merges both impossibilities into the single word "uncountable". The Z-optics splits them: the first fits into Z entirely (it is a property of extensional presentation and strikes even countable stream families), the second remains a cardinality fact which the diagonal renders *earned*. Resonance: the split "how many / which exactly" of §12 (cardinality earned without identity) is the same split seen sideways.

Kernel-checked (`ZExped.lean`, zero axioms). Both failures are theorems. The stream atom never earns T at any time (`eqStream_never_T`), not even a stream against itself (`eqStream_self`); one finite witness earns apartness (`eqStream_apart`) and earned apartness is never revoked (`eqStream_F_persist`). Failure #1: no registry certifies membership of anything, its own rows included (`mem_never_T`). Failure #2: the diagonal earns its non-membership against every entry (`diag_not_member`). What is *not* formalised is the contrast case — that registries of fractions certify every element — so the countability of $\mathbb{Q}$ as earnable presentation identity remains measured.

14. Functions: taint mode (MEASURED + Lean)

A function is a computation, not a verdict ⇒ by the two-register theorem it behaves lazily: **the mark flows through the function** with a growing pedigree (f(m) is a new mark "f applied to m"). Measured:

- **Images:** verified collisions *earn* merging (f(1)=f(2) is a proven fact, the core deduplicates), marks keep multiplicity: $|f(\{1,2,3,Z\})| \in [2,3]$.
- **Composition:** taint is transitive (pedigree g(f(m))); the image is associative at representation level while verdict-equality is F (regularity R1, §26).
- **The preimage splits** into a verdict version (marks dropped — default deny) and a solver version (marks as candidates) — regularity R2 (§26).
- **The pearl: even the identity function is not certifiably injective on a marked domain** — pairs with a mark give Z-atoms, the implication Z→Z = F, the ∀-fold collapses. An injectivity certificate requires a fully verified domain; the echo of fallen $S \subseteq S$.
- **Laundering is forbidden:** functions do not remove marks; the only sanitizer is external verification of the value. In security terms: declassification only through proof.

The third engineering twin. After IEEE NaN and SQL NULL — **taint tracking / information flow control** (Denning's lattice 1976, Perl taint mode, TaintDroid): the Z-mark is taint, lazy flow through computations is taint propagation, greedy verdicts are sanitizer checks, the laundering ban is no-declassification.

Kernel-checked, and now in two places. The pearl is a theorem: `ZExped.inj_cert_marked` shows that ONE marked pair collapses the injectivity certificate for EVERY function — the identity included — resting on `eqAtom_z_right` (an atom against a mark is Z). And the laundering ban is a theorem too (`ZTaint.no_laundering`; and the tradition's own model is embedded in `ZFlow.lean`, §27): applied to an unverified reference, a chain of verified functions OF ANY LENGTH returns an unverified reference. The pedigree grows; the mark never comes off. The sanitizer is not an application at all — a verified value is a different ELEMENT, put there by an act of checking, and that is stated beside it. In security terms: no declassification without proof, proved.

The rest of this section — images and multiplicity, transitivity of the pedigree at representation level, the preimage split, the merge of verified collisions — is measured on worked cases and is **not** formalised. The tag stays MEASURED for that reason.

15. Arithmetic with marks (MEASURED + Lean)

Numbers: verified values and marks with an interval of partial knowledge [lo,hi] (ignorance = $(-\infty,\infty)$). Operations are computations ⇒ lazy: intervals flow (interval arithmetic, decorrelated). Comparison atoms follow the generating principle extended to intervals: **T if forced under all readings; F if falsehood is forced; else Z**. Measured:

- **Forcedness earns even on marks:** $0\cdot w$ = an earned 0 even for a wild mark (forced on $\mathbb{Z}$ by all readings) — a point of deliberate divergence from IEEE (their 0·NaN = NaN: their domain contains inf/nan) — now a theorem, `ZNaN.zero_times_mark` / `emb_mul_not_hom`, §27. $m-m \in [-9,9] \neq 0$ — decorrelation (like NaN−NaN, like {Z,Z}).
- **Three fates of an atom:** [3,5]<[10,12] — T earned; [3,5]=[10,12] — **apartness earned by intervals** (the echo of §13: difference is finitely witnessable); overlap — Z; the same mark against itself — Z (coincidence of bounds ≠ coincidence: identity is earned by nothing short of full verification [x,x]).
- **Verification = interval narrowing:** the atom "4 < m" travels Z → Z → T along [0,9]→[3,7]→[5,7]; what is earned is never revoked — the monotonicity of the lazy register, now in numbers.
- **Price-list inheritance:** commutativity of addition survives at the interval level, verdict-equality is Z→F; the unit x+0=x falls verdict-wise with coinciding intervals (regularity R1, §26) — now a theorem in both halves (`ZNumPrice.lean`, E62): the reading sets of x+y and y+x, and of x+0 and x, coincide (`add_comm_readings`, `add_zero_readings`), and on a mark with two readings neither equation is forced true or forced false (`comm_not_earned`, `unit_not_earned`; the cell above, $x\in[1,3]$, $y\in[2,4]$, is `zarith_instance`).

The fourth twin: abstract interpretation (Cousot & Cousot, 1977) — interval value analysis (lazy flow of abstract values through computations) + assertion checking (greedy verdicts). Of the four engineering traditions named so far — NaN, NULL, taint tracking, abstract interpretation — all four are now embeddings (`ZNaN.lean`, `ZNull.lean`, `ZFlow.lean`, §27; and this one, below).

Kernel-checked embedding (`ZAbsInt.lean`, empty axiom list). Their Galois connection is formalised as they state it — $\alpha(S)$ = the least interval containing the value set, γ the concretization — and proved in both directions: $\alpha(S) \sqsubseteq [a,b] \iff S \subseteq \gamma([a,b])$. Against it our verdict is not merely *sound* on the abstract value, which is what abstract interpretation normally buys and pays precision for; it is **exact**. For a threshold atom, the verdict computed from the interval alone equals the verdict computed over the whole concrete value set, in all three cells. Nothing is paid.

And the price appears exactly where this section already said it does. The moment one variable occurs twice the exactness is gone: over the value set [1,3] every concrete `v - v` is 0, so the concrete verdict of `v - v < 1` is T, while the decorrelated interval computation gives [0,2] and returns Z. The witness is mechanical, not narrated — the concrete image and the abstract interval are both computed in the file. So decorrelation is not an aside about intervals; it is the exact boundary of the exactness theorem.

What is not done: the framework — widening, narrowing, fixpoint transfer — is not formalised. One atom over one abstract domain is mapped, not the method.

Kernel-checked (`ZExped.lean`, `ZNum.lean`, `ZNumCoherent.lean`, zero axioms). Identity is earned by nothing short of full verification: a mark against itself is Z even when the bounds coincide (`mark_self_not_earned`).

Narrowing-heredity — what is earned is never revoked as intervals shrink — is a theorem for every comparison atom under both readings (`forcedLE/NotLE/LT/NotLT/EQ/NE_hereditary`, twice over), together with the transitivity of narrowing and the endpoint case. Decorrelation has a named witness (`decorrelation_witness`). Forcedness on products (0·w) is `ZNaN.zero_times_mark` (§27), and the price-list inheritance for commutativity and the unit is `ZNumPrice.lean`.

16. The probabilistic bridge: Z ≠ p = 0.5 (MEASURED + Lean)

Three measurements answer how a mark of ignorance differs from a uniform prior.

Reparametrization (a discrete Bertrand). Given only $w \in [0,1]$; the question "$w \le 0.25$?". A Bayesian with a uniform prior on w answers 1/4; a Bayesian with a uniform prior on w^2 (the same ignorance!) answers 1/16. One ignorance — two contradicting numbers: the choice of parametrization is smuggled information. The ZTL atom at $w \in [0,1]$ is Z in both parametrizations: **ignorance does not convert into a number without importing information**, and this is invariant.

Dempster–Shafer. On masses m({a})=m({a,b,c})=1/2 it is measured that the ZTL verdict of an event is the threshold of belief functions: **T ⇔ Bel = 1** (forced by all readings of ignorance), **F ⇔ Pl = 0** (excluded by all), else Z. The generating principle of ZTL is the {0,1}-threshold of Dempster–Shafer theory; the interval cardinality of §12 is the elementwise [Bel-count, Pl-count].

Ellsberg. An urn with a verified 50/50 versus an urn of unknown composition: EV(K)=[50,50] is an earned number, EV(U)=[0,100] an interval; the atom "U is no worse than K" is not forced → default deny → choose K. The famous "irrationality" of Ellsberg's subjects (1961) is the distinction between risk (verified p) and ignorance (a mark), inaccessible to a point prior: zero-trust rationality.

The second "SQL theorem" — about Bayesians. A point prior made from ignorance is a greedy laundering of Z into a number, the same substitution as merging NULLs in DISTINCT, and it is punishable (reparametrization). The honest architecture is two-registered: ignorance lives in mass intervals (the lazy register — Dempster–Shafer and **the fifth twin: Walley's imprecise probabilities, 1991**), while decisions are verdicts by forcedness (the greedy one). Bayes remains honest on verified probabilities — his C-extension; only minting numbers out of emptiness is forbidden.

The Dempster–Shafer bridge is now a theorem (`ZDempster.lean`, empty axiom list). Their side is formalised as they define it — a mass assignment is a list of focal subsets of a finite frame with positive weights, Bel(A) sums the focals lying inside A, Pl(A) those meeting it. Our side is defined by the generating principle alone, with neither Bel nor Pl appearing in it: forced true when every reading of the ignorance lands in A, forced false when none can, marked otherwise. The two are then proved to agree in all three cells, for every finite frame, every proper mass assignment and every event — so what was measured on one assignment holds on all of them. With it the fifth twin (Walley's imprecise probabilities, whose lower/upper pair is exactly Bel/Pl) joins the sixth as an embedding rather than a correspondence.

The properness condition is not decoration, and finding that out is what the theorem was for. The first draft required only positive weights and was FALSE: on the empty mass assignment the verdict comes out T vacuously while Pl = 0, and the F-cell fails; an empty focal breaks it the same way. That is Dempster–Shafer's own `m(∅) = 0`, which held silently on the one assignment that had been measured. A single checked instance cannot show you the condition it happens to satisfy.

And the other two claims are now theorems as well (`ZIgnorance.lean`, empty axiom list).

Reparametrization. The two Bayesian numbers are computed and shown different — by cross-multiplication, so that no division enters a file about not importing what one was not given. Our verdict is Z in both, and not by luck: it is proved invariant under EVERY strictly monotone relabelling, not only under the squaring the example uses. A monotone map carries the gap to a gap; the number moves and the verdict does not.

Ellsberg. The verified urn earns the claim "no worse than fifty-fifty" and the unknown urn does not — and the module proves WHY it does not, which is the whole content: the atom itself is Z, neither forced nor excluded, and the denial of an unforced claim is classical (`¬Z = F`, §3.1). Default deny, with no judgement passed on the unknown urn's merits. What Ellsberg's subjects were reading is a mark, and a point prior has no place to put one.

17. The modal layer: local □ versus global (MEASURED + Lean)

Worlds are the classical completions of the unverified atoms; □φ = in all, ◇φ = in at least one. Measured:

- **Atom verdicts are modal thresholds** (totally): T ⟺ □p, F ⟺ □¬p, Z ⟺ contingency; the duality ◇ = ¬□¬ holds; nested modality collapses (□p is classical ⇒ □□=□) — the frame is S5-like. This is §16 one floor up: □/◇ are the Bel/Pl thresholds in probabilistic dress.
- **The tableau signs are modal claims**: strict T/F are □φ and □¬φ, weak P/N are ◇φ and ◇¬φ. The calculus signature reads modally: proof demands necessity, refutation settles for possibility.
- **Three logics on one formula** (classical | global □ | ZTL): p→p and LEM: T | T | F — supervaluation (one □ over the whole formula) preserves all classical tautologies, the local per-operator □ fells them. But ¬¬p: T | Z | T — **ZTL earns a verdict which global supervaluation cannot give** (the ladder of floors): the systems are incomparable, not ordered by strictness.

Result: **ZTL is a locally-modal logic over the S5 frame of completions**; every operator carries its own □-collapse. Bochvar's translations (→ = ◇A⊃□B, ¬ = □¬) acquire world semantics; the supervaluation/ZTL split turns out to be the global/local modality split. The theoretical relative is Hintikka's epistemic S5 (□ = "known"): ZTL asserts only the known, but its modality is per-operator.

Kernel-checked (`ZExped.lean`, zero axioms). The threshold reading is a theorem in all three directions at once — T ⟺ □, F ⟺ ¬◇, Z ⟺ contingency (`atom_thresholds`) — with the duality ◇ = ¬□¬ over completions (`box_dia_duality`). And the separating cell is machine-checked (`ladder_vs_global`): the local ladder earns ¬¬Z = T exactly where the global □ goes mute, so the two layers are incomparable rather than one refining the other.

18. Russell: containment instead of explosion (MEASURED + Lean)

Russell is the liar dressed in membership: R = {x : x∉x} ⇒ R∈R ⟺ ¬(R∈R). The test universe: a = ∅, b = {b} (a lawful eccentric), R; a system of nine membership facts, with Russell's definition referring to the facts x∈x. Measured:

- **The greedy jump**: zero models, iteration of period 2, and exactly one cell oscillates — R∈R; the whole storm is localized.
- **Lazy grounding**: 8 of 9 facts grounded; a∈R = T (does not contain itself — admitted), b∈R = F (contains itself — rejected). **Russell's set works as a set for everyone except itself**; quarantine is one cell.
- **Verdicts**: "R∈R?" — not earned, refusal; "R∉R?" — the greedy reading of ¬(R∈R) is F — also refusal. The NaN signature reaches set theory: neither membership nor non-membership is earned.
- **The twin S = {x : x∈x}**: three greedy models (T, F, Z), lazy grounding Z — the truth-teller of set theory, underdetermination. Even and odd, now in sets.

The contrast with the classical outcome is maximal: for Frege one cell R∈R blew up the whole system (from an asserted contradiction everything follows). In ZTL the same cell goes into quarantine and the rest of the universe stays grounded: **explosion requires an asserted contradiction, and quarantine asserts nothing**. The grounding is kernel-checked: the certified least-fixed-point iteration of §9 computes the nine-fact system to [F,F,T,F,T,F,F,F,Z] — eight facts classical, quarantine exactly at R∈R (Lean `russell_grounded`, zero axioms). A comparison with an earlier system of the author is instructive: in VR-Sets [Zenodo 10.5281/zenodo.20592428] Russell is excluded by grammar (forbidden to write), in ZTL he is admitted and defused pointwise. Two honest answers to one calamity: keep it out, or let it in under guard.

19. The verification operation and verdict warranties (MEASURED + Lean)

The act of verification — removing a mark and writing in the earned value — exposes a narrow place: **greedy verdicts are non-monotone under verification**. Not only refusals flip (p∨¬p: F → T upon p:=T — expected: default deny until checked) but **T flips too**: ¬¬p = T (a ladder report) dies at p:=F. A verdict without a warranty is a Frege cell: an unfenced spot where a consumer who read T as "settled forever" builds on sand.

The fence is a warranty — and (corrected in v1.1) it is a ladder of two grades, not one bit.

- **The SOUND grade** (the stability bit of v1.0): all completions give one classical answer equal to the current greedy verdict — the global supervaluation of §17. It buys *never lies*: a sound verdict agrees with every possible resolution of the marks, so no truthful verification can ever reveal it to have been false. Cheap: one pass over the completions.
- **The HEREDITARY grade**: the verdict is unchanged under *every partial refinement* — any subset of the marks verified to any classical values. It buys *never spoils*: no verification path can revoke the verdict. Hereditary ⟹ sound (completions are among the refinements); the converse is **false**.

Correction of the v1.0 claim (MEASURED). v1.0 asserted an equivalence — "stability-by-supervaluation ⟺ invariance under every sequence of verifications (totally, 90 formula×marking pairs)" — and a monotonicity corollary. Both were facts about that 10-formula pool, not laws. The separating shape, found by the identity atoms of the operational-sets expedition (E21) on a 3303-formula pool and cross-checked with this section's own instruments, is **or(ladder, gap)**:

> ¬¬p ∨ (q ∨ ¬q) — greedy T via the ¬¬ ladder, insured by a gap that is true in ALL completions yet greedy-F; verifying p:=F kicks the ladder away before the gap closes. The verdict was sound — and died. A simpler cell: (¬p) → (q→q), where the gap is the fallen law of identity itself.

Measured on the extended pool: hereditary-without-sound 0 (the ladder is real), sound-without-hereditary > 0 (the grades separate); the hereditary grade is monotone — never revoked, never degraded (0 violations, totally) — while sound-only verdicts are revocable. Subsequently stress-tested at scale (2026-07-12): a 151.8-million-pair hunt over four atoms at depth three, all six connectives, found 0 violations of the ladder inclusion and of hereditary monotonicity; the grade separation is generic (1.2 million cells), not exotic.

The fence depth is exactly m−1 (MEASURED + Lean). How deep must a verification-invariance check look before the hereditary grade is certain? For a *sound* verdict all full completions agree with it by definition, so heredity violations can live only at partial refinements of size at most m−1 (m the number of marks): depth m−1 always suffices. This half is now a theorem too (`ZFenceDepth.resolved_all_marks_agrees`, empty axiom list): a refinement that has resolved every mark the formula *depends on* has become an ending, and endings agree by soundness. It is stated positively — "resolved everything relevant ⟹ agrees" — rather than as the existential "a violation leaves a mark unresolved", because the latter is an ∃ drawn from a negation and would have to come through `Classical.byContradiction`, taking the whole file to the classical tier for a cosmetic gain. The content is the same, read by contraposition. It is also necessary: the guard family

```
(b₁ ∧ … ∧ b_{m−1}) → (a → a)
```

— a conjunction guard of m−1 marks standing over the fallen law of identity — is sound, invariant under every verification of fewer than m−1 atoms, and dies the moment all guards are verified true, when the door opens onto the greedy-F gap a→a. This was checked deterministically for m = 3, 4, 5, with the m = 2 witness (¬p)→(q→q) above; **it is now a theorem for every m at once** (`ZFenceDepth.lean`, empty axiom list), because the guard family is uniform in m and so generalises whole. Three clauses are proved: the cell is sound at every m; it is invariant under every refinement leaving any single guard unverified — and that clause came out *stronger* than the argument needed, since a sunk guard makes the arrow vacuously true whatever happened to the gap atom; and it dies exactly when the full guard set is verified. The violation is therefore not merely reachable at depth m−1, it is reachable nowhere else. One step was left as prose at E33 rather than smuggled into the formalisation — that a check inspecting fewer than m−1 atoms must leave some guard unverified is finite counting, not logic — and E58 later proved it (`length_ge_of_all_below`, below). Hence **no constant-depth characterization of the hereditary grade exists**; the cost of the full warranty grows with the number of unverified inputs.

And no structural, non-enumerative criterion exists either — the check IS a tautology check (E57, `ZHeredTaut.lean`, empty axiom list). The fence family read the other way: put a classical formula ψ in the gap and make the guard the *meter of verification* — excluded middle, which is not a law here, reads `Z ∨ ¬Z = F` at a mark and T on any verified value, so `G = ∧ᵢ (xᵢ ∨ ¬xᵢ)` is F while any guarded atom is unverified and T once all are. The witness `G → ψ` reads T at the all-marked start (F → _ = T), stays T under every refinement that leaves a guard marked, and reads `T → ψ = ψ` at every full completion. So its T verdict is hereditary exactly when ψ is true at every classical point: `hereditary_iff_taut` — heredity of the witness ⟺ tautology-hood of ψ, for every ψ over the guarded atoms. The witness has the size of ψ plus the guard, so a procedure deciding the hereditary grade in time polynomial in the formula would decide TAUT: the grade is coNP-hard, and a cheap exact criterion would put coNP in P. (Membership in coNP — a revoking refinement as a certificate, one greedy pass — is argued in prose and not formalised, so the claim stops at hardness; "coNP-complete" is not claimed.) What exists are SUFFICIENT structural conditions — `NoGift.no_gift`, no mark under a negation — and the reduction says the gap between sufficient and exact is not one more theorem away but a complexity class. MEASURED (`zheredtaut.py`): on all 2928 formulas of depth ≤ 2 over two atoms as the generator enumerates them (2906 distinct — it lists 22 depth-1 formulas twice, which moves no verdict; 588 tautologies) and 2000 random formulas of depth ≤ 3 over three atoms (171 tautologies), the brute-force hereditary grade of `zverify.py` and classical tautology-hood agree on every one — zero divergences.

Result: a verdict is a pair (value, warranty grade). The value is greedy (local, fast); the warranty grades are global. Six verdict classes now, and the honest advice differs by grade: **hereditary T — build your house** (no verification path can revoke it); **sound T — never a lie, but may stall to refusal before verification completes**; T-until-verification — a ladder report (¬¬p), alive till the first check; symmetrically for F, with F-until-verification as default deny. The Frege cell is fenced by the top grade only; the middle grade fences lying, not spoiling.

A verdict carries a warranty; a refusal carries a receipt (Lean, `Receipt.lean` and `LabelExact.lean`, zero axioms). The lazy register answers Z together with a label — which unverified atoms the answer is still waiting on. That label is what makes a refusal accountable rather than merely withheld, and it is now bounded from both sides.

Nothing that could matter is omitted (`receipt_complete`): for every formula of the language, every valuation and every unverified atom, if the label leaves that atom out then no value of it changes the answer. Contrapositively — the reading that matters — **an atom that could change the answer is always named.** The same holds in the register that actually issues verdicts (`receipt_complete_greedy`). That second theorem was predicted in writing to be FALSE, by us, before the census, on the ground that the greedy lift is non-monotone and connective-local; the census refuted the prediction (zero misses in 296,161 cells carrying an unverified atom) and the reason turned out to be `greedy_agrees_when_decided` plus the shape of the greedy tables. The prediction is recorded rather than dropped.

And nothing idle is named — on a linear claim (`label_exact_linear`): if no unverified atom is read twice, every atom on the receipt is one the answer is genuinely waiting on — there is a definite reading of the other unverified atoms under which answering this one T and answering it F give different verdicts. Two theorems carry that sentence. The kernel's `pivotal` lets the reading of the others stay partial, and `label_exact_linear` is proved with it; the definite form is `label_exact_linear_definite` (`LabelExactDefinite.lean`, E60), proved over the partial one rather than by a second induction: a side that reads Z is a pending linear claim at a marking of its own, so `drivable` pushes it to the value the other side does not have, the definite side stays where it was by the monotonicity of the lazy register, and the remaining marks inside the claim are filled. (For a day this sentence said "definite" with only the partial theorem behind it; E59 found the gap, softened the sentence, measured the strong form — 57,969 named atoms on 48,759 pending linear cells, no failure — and E60 proved it. Outside the linear class the same census finds 56,370 named atoms of 153,522 that no definite reading moves, `p ∧ ¬p` first among them.)

Three things this pair does not say, each of them measured.

- **Relevance cannot be tested one atom at a time.** `p ⊕ q` is linear, both atoms are named correctly, and neither moves the answer alone (kxor Z Z = kxor T Z = kxor F Z = Z). Single-atom probing — the notion we first measured against, which produced our own inflated over-approximation figures of 16% at depth 2 and 35% at depth 6 — systematically reports jointly relevant atoms as idle. Against the joint notion (the union of minimal jointly moving sets) the label is exact in 93% of 14,530 pending cells and never once too small.
- **The linearity hypothesis is load-bearing, not decorative**, and this is proved rather than asserted: `clash_names_an_idle_atom` exhibits `p ∧ ¬p`, where the receipt names an atom that no reading can move. All of the residual inexactness (7% — 994 of the 14,530 pending cells above, `lab/label/bench.py`) sits in cells where a marked atom occurs more than once — the same occurrence-independence that costs `p → p` and `p ∨ ¬p`.
- **The residue was attacked and did not yield.** A demand-driven label computed against reachable sibling values was built as a prototype and matched the existing label cell for cell — the existing analysis already IS that analysis. Closing the gap needs relational tracking of shared occurrences across branches, which is not cheap.

And the residue cannot be closed cheaply at all — the exact receipt is NP-hard (E59, `ZReceiptHard.lean`, empty axiom list). Put a formula ψ that does not read `a` beside the false that keeps the mark, `ψ ∧ ¬ψ` — F wherever ψ is settled, Z wherever it still waits — in the witness `(a ∨ (ψ ∧ ¬ψ)) ∧ ψ`. At the all-marked start the receipt names `a` (`named`), by its own rule, both branches pending; and `a` was worth naming exactly when ψ has a classical point (`pivotal_iff_sat`): with `a := T` the claim reads ψ, with `a := F` it reads `ψ ∧ ¬ψ`, the two differ precisely where ψ reads T, and a partial reading on which ψ reads T lies below a classical one by the monotonicity of the lazy register. So deciding whether a named atom is idle is deciding SAT(ψ): the exact receipt is NP-hard (membership in NP — a pair of readings as a certificate — is argued in prose and not formalised, so the claim stops at hardness and "NP-complete" is not claimed). The simpler `a ∧ ψ` is no reduction, and the kernel says so (`conj_always_pivotal`): under a reading that leaves ψ waiting, `a := T` gives Z and `a := F` gives F, so `a` is pivotal whatever ψ is — the false that keeps the mark is what makes both sides wait together. Hence `labF`, complete and exact on linear claims, is the forced cut, as `joint` is for the width: with E57 and E58, the three grades the judge does not compute are the three it cannot compute cheaply — hereditary (coNP-hard), exact width (NP-hard), exact receipt (NP-hard). MEASURED (`zreceipthard.py`, the judge's own lazy evaluator, and `pivotal` read literally, partial readings included): on all 2906 formulas ψ of depth ≤ 2 over two atoms and 600 random formulas of depth ≤ 3 over three, `a` is named on every one, "pivotal $\Longleftrightarrow$ satisfiable" holds on every one, and `a ∧ ψ` is pivotal on every one — zero divergences.

Two auxiliaries carry the result. `drivable`: a pending linear claim can be driven to T and to F (48,759 pending linear cells, every one reachable both ways) — so a refusal never hides a foregone conclusion; this is also what makes the exactness proof go through, since holding a pending sibling at Z would hide the pivot. And `Linear.linear_no_loss`, from the other direction: at multiplicity ≤ 1 no truth is lost. Together, **on a linear claim the judge is exact in both directions — it loses no truth and names no idle ground.** What remains imperfect there is the over-grant, and that one comes from the collapse $\neg Z = F$ rather than from multiplicity (`¬¬p` has one occurrence and still grants).

And the over-grant has its own fence (Lean, `NoGift.lean`, zero axioms). The gift is the dangerous half — an unearned T is indistinguishable from an earned one, where a refusal at least announces itself — so the question is which claims cannot produce one. The answer is an axis orthogonal to multiplicity: **negation.** If every unverified atom of a claim stands under no negation (`ContextClosure.negFree` for each mark — which also bars it from an implication's antecedent and from ⊕/↔ entirely), then a greedy T survives every refinement of the marks:

```
no_gift : posMarks v ψ → refines v w →
          evalF v ψ = T → evalF w ψ = T
```

`closure_coincides` was the one-atom case; what is added is all marks at once and all PARTIAL refinements, which is what the hereditary grade actually asks for.

Three things must be said with it, and each is proved rather than asserted. **Only T is protected** (`F_is_not_protected`): `p ∧ p` sits inside the fragment, is greedily F, and revives at `p := T` — measured, 950 of 1700 in-fragment F cells are revocable. The asymmetry is the point: a refusal that later becomes a verdict is inquiry working. **The fragment is sufficient and narrow** (`fragment_is_not_necessary`): `p → q` with `p` unverified and `q` verified true cannot be moved by any refinement, and sits outside — and outside is where most safe verdicts live, 66% of hereditary verdicts at depth 2, rising to 99% at depth 6. Outside is unguaranteed, not unsafe. **And the two fragments are incomparable**: gifts do not need multiplicity (594 linear cells gift at depth 2) and in-fragment claims may read a mark twice without gifting (184 such cells). Negation is the axis of the gift; multiplicity is the axis of the loss.

Measured before the proof was attempted, predictions frozen in `lab/nogift/PREDICTIONS.md`: 323,530 cells over three pools, zero gifts inside the fragment; all five predictions held, including the one recorded as the weakest.

Measured before each proof was attempted, in the order the method requires: 412,593 cells for completeness, 14,530 pending cells for the joint-exactness figure, 128,372 linear cells for exactness (zero inexact), 48,759 for drivability. Label propagation of this kind is old — de Kleer's ATMS [40] — and is named here rather than presented as new; what is ours is the pair of bounds on this particular label.

What the receipt cannot answer: how many at once (MEASURED). The label says which grounds could matter. It never says how many must be settled *together* before anything moves, and the difference is operational. Define, for an unsettled claim, its **width**: the size of the smallest set of unverified grounds that, filled together, moves the verdict. Width 1 is ordinary incremental inquiry — go check this one. Width ≥ 2 means no partial progress is possible: the whole configuration must be produced before anything at all happens.

Width 1 dominates, and the judge's one-ground order is usually honest: 93% of unsettled cells over the exhaustive depth-2 pool, 91% over random depth-5 formulas across five atoms. **But width is not bounded.** The hunt found width reaching the number of available grounds — 4 of 4 at four atoms, 5 of 5 at five, over ~24,000 unsettled cells. The honest statement is a limit rather than a defect:

> **Step-by-step inquiry is not always possible.** There are claims of any size for which nothing moves until every unverified ground is filled at once.

That is Meno's second horn in a measurable form, and it is measured narrowly: the claim is about this calculus's own pool, not about inquiry in general.

The prediction behind this was right in its conclusion and wrong in its witness, which is worth recording. We predicted unbounded width with xor chains as the witnesses, having worked `p ⊕ q ⊕ r` by hand. That formula has width **1** — the inner xor collapses to a definite value and the outer one is sensitive to the last ground alone; the hand-calculation had tested two atoms of three. The genuinely wide cases are irregular mixed formulas, and they had to be hunted rather than constructed.

Width is invisible to the label, and this is the over-approximation half of §19 showing its operational face: `p ⊕ q ⊕ r` names all three grounds and has width 1; `p ⊕ q` names both and has width 2. So the fix could not be a better label — it had to be a new computation. The judge now reports a `joint` field: when no single ground moves the matter, it names the grounds that must be filled together and says plainly that a one-at-a-time order would be empty work. The exact width is **deliberately not computed** — that search is exponential in the number of marks, and what a reader needs is the difference between "go check this" and "no single check will move this", not the cardinality.

And the restraint is forced, not chosen (E58, `ZWidthHard.lean`, empty axiom list). On E57's witness $(\wedge_i\ x_i \vee \neg x_i) \to \psi$ at the all-marked start, nothing short of the whole guard moves the verdict — a refinement that leaves any guarded atom marked keeps the guard F and the verdict T, so every moving set contains all n+1 guarded atoms (`no_move_below`, with the counting step E33 left as prose now proved: `length_ge_of_all_below`) — and the whole guard moves it exactly when ψ has a falsifying classical point (`width_iff_refutable`). The width of the witness is therefore n+1 or nothing, split precisely by whether ¬ψ is satisfiable: "is the width ≤ k?" is NP-hard (the witness carries SAT(¬ψ); that a moving set with its values is a certificate, one greedy pass, would give membership in NP — argued in prose, not formalised, so the claim stops at hardness and "NP-complete" is not claimed). Width 1 is cheap — 2m evaluations, which is what `joint` does — and the exact width cannot be, unless P = NP. Together with E57 (no width at all ⟺ hereditary ⟺ a tautology check) and E59 above, the three grades the judge does not compute are the three it cannot: one coNP-hard, two NP-hard. MEASURED (`zwidthhard.py`, the instrument's own `width`): on the same 2928-formula enumeration of depth ≤ 2 over two atoms the witness's width is 2 on the 2340 non-tautologies and undefined on the 588 tautologies; on 600 random formulas of depth ≤ 3 over three atoms, 3 on 544 and undefined on 56 — zero divergences from the theorem.

20. Evidence combination: conflict is not laundered (MEASURED + Lean)

Pieces of evidence about one value are constraints; **combination = intersection**. Measured:

- **The unification result** (a theorem for interval constraints, `ZCombine.lean`; measured beyond them): verify of §19 is a special case of combination (a singleton witness [v,v]); verification and evidence fusion are one operation. An honest side effect: the act of checking can itself earn a conflict (verify 7 against m∈[0,5] — the checker against the prior evidence).
- **An empty intersection is an earned contradiction of sources** (a sound F for the verdict "both are honest"), not noise for renormalization.
- **Zadeh's paradox (1984), resolved in Smets' favor.** Two doctors almost exclude a tumor (0.01 each); Dempster's rule renormalizes the 0.9999 conflict away and outputs "tumor = 1" — an unshakable false certainty. Smets' conjunctive rule (TBM) keeps the conflict in m(∅) — our approach: the conflict is exhibited, the diagnostic verdict is refusal until the doctors are sorted out. **Renormalizing conflict is the same laundering of ignorance as the uniform prior (§16), in the chapter on combination: one principle, two diseases.**
- **The sixth twin: provenance polynomials (Green–Karvounarakis–Tannen, 2007).** The pedigrees of marks (§14) grow into polynomials over sources: a fact with derivations A·B + C stays alive while at least one monomial lives; retracting a source zeroes a variable. An algebra of trust in derivations, measured on retraction scenarios.

The twin count: **six** — NaN, NULL, taint/IFC, abstract interpretation, imprecise probabilities, semiring provenance.

The sixth twin is now an embedding; this section's own claims are kernel-checked below. `ZProv.lean` (empty axiom list) formalises the provenance algebra and maps it into ZTL — see §27 for what that theorem says and, more to the point, what it does not. It settles the twin, not this section. The section's own three claims — combination as intersection, the earned contradiction of an empty intersection, and the Zadeh case — were measured on worked scenarios with `zcombine` first and are kernel-checked below (`ZCombine.lean`, E39); the tag stays MEASURED + Lean for what the closing paragraph names as measured only. One further thing belongs here and is not comfortable: our own probe against the installed package (`db/probe_provenance.py`, PostgreSQL 16.14 / ProvSQL 1.13.0-dev) found that semirings already do the cascade, the alternatives and the exposed set, and that ProvSQL also carries magnitudes through aggregation — which an earlier version of this corpus denied in print. That denial is withdrawn. Anyone needing those four things should use ProvSQL; what this paper adds is where their algebra sits inside the two registers, and where it provably cannot.

Kernel-checked (`ZCombine.lean`, empty axiom list). Combination is the meet of constraints and its members are exactly the values both sources allow; an empty meet is an EARNED contradiction in the sense that makes the word mean something — no value whatever satisfies both, so the refutation holds under every reading; and verification is that same operation against a singleton, which is why the act of checking can itself earn a conflict when the checked value lies outside what was already known. One operation, proved, not two.

And Zadeh's paradox is now a theorem rather than an anecdote. Two doctors, three diagnoses, each giving the tumour one part in a hundred and disagreeing about the rest. Dempster's rule intersects the focal elements, discards what intersects to nothing — 9999 parts of 10000 — and what survives sits on the tumour alone. By the threshold theorem proved for their own theory in §16, the verdict is then T and belief is full: an unshakable diagnosis manufactured from two one-percent opinions. *The certainty is derived inside our machinery, from their rule, using our theorem about their theory* — not criticised from outside. Retaining the conflict instead puts the mass on the empty focal, and the assignment is then improper: no verdict is issued and the conflict stays visible. That is the refusal this section describes, and it is a theorem about the precondition rather than a further verdict.

Still measured, not proved: the unification of verify with combination is proved above for interval constraints only, and the provenance-polynomial bullet is covered separately by `ZProv.lean` (§27). The Smets/TBM reading of `m(∅)` as a modelling decision rather than an error is a position, not a theorem, and is not claimed as one.

21. Logical time: verification is the only clock (MEASURED + Lean)

ZTL owns no physical clock, and §7 keeps it honest: duration, tense and "how long" are outside the system's axis. Yet one clock was inside the calculus all along, unnamed. The verification operation of §19 is a tick: one act `verify` resolves one mark into an earned classical value. A moment is a marking; the past is the verified prefix; the future is a tree — every remaining mark can resolve either way, so time branches, and "everyone's time differs" is just the choice of a path through one tree.

The ladder is a temporal logic. The central identification costs nothing and buys the whole layer: the warranty ladder of §19, read on the verification tree, is a system of temporal quantifiers —

- *until-verification* — true **now** (the present tense of a verdict; credit);
- *sound* — true **at every ending** (all completed traces agree; the road may wobble);
- *hereditary* — true **always along every path** (the invariant of the tree; never revoked).

The modal layer of §17 is the statics of this structure (worlds = completions); the temporal layer is its dynamics — the paths through the partial refinements between here and the endings.

Measured (`ztime.py`; the grade automaton over the exhaustive depth-≤2 pool, 2,906 formulas, 29,812 ticks): the hereditary states absorb — not one tick leaves them (0 violations); no compound formula is ever caught waiting (greediness in temporal costume); ground can arrive all at once (14,818 direct U→H jumps) and credit can worsen before it settles (108 S→U demotions); every completed trace ends hereditary — the arrow of logical time points at the shelf (130 of 130); and settlement can come early — 68 of 130 traces reach a hereditary verdict with marks still unresolved (after `p:=T` the verdict of p∨q no longer cares about q). Settling times run from 0 (¬(p∧¬p) is born on the shelf) to every-mark (⊕ needs them all).

Lean (`lean/ZTime.lean`, empty axiom list, structural — for every formula, marking and tick, not an enumeration): absorption (a hereditary verdict survives any tick with its grade), the arrow (a fully verified marking is hereditary), and the ladder inclusion (hereditary ⟹ sound; an ending is a refinement). The proofs are pointwise throughout — no function extensionality enters.

A conjecture that lived one hour — kept, as the method demands. The first sweeps found no tick that ever *enters* the sound-only grade, and for an hour the conjecture stood: *sound is a birth grade — verification spends it but cannot mint it*. The proof attempt broke at a specific spot, and the spot folded into a counterexample: the pools had been shallow. The selector $\varphi = (a \wedge X) \vee (\neg a \wedge p)$ with $X = \neg\neg p \vee (q \vee \neg q)$ walks F/U → (a:=T) → T/S → (p:=T) → T/H from the all-marked start: soundness is *earned* — by the tick that verifies which world you are in — and the full strict ladder U→S→H is realized rung by rung (`ztime.py` §§5–6, the record of the death included).

The tool. The ZFL statement genre carries an optional `timeline` field (a list of verification ticks, validated with its own error codes); the engine plays it into a chronicle — verdict and warranty per tick, with settlement marked. The consumer sentence of the layer: once a verdict is hereditary, every remaining check buys nothing — stop paying (`usage/car.py`, the used-car stand: an affirmation is earned at the last tick; a refusal is grounded at the first failure, three checks saved; verifying the selector first saves two).

22. Epochs: expiry and the boundary theorem (MEASURED + Lean)

The time of §21 is monotone: ground only arrives. Institutions live in non-monotone time — confirmations lapse, registries are re-pledged, facts are authoritatively changed. The anti-tick `expire` returns an earned value to the mark, and with it the layer splits into two chronologies that must never be conflated: the *knowledge* chronology (verify — learning more about the same world) and the *validity* chronology (expire — the world becoming different). A maximal verify-only stretch is an **epoch**; an expiry opens a new one. A composite event — authoritative revocation — is deliberately a composition, expire + verify of the new value, so that one event cannot hide two institutionally different facts.

Measured (`zexpire.py`): hereditary is a warranty against future *verification*, not against the *loss of ground* — a fully verified conjunction (T/hereditary) falls to F/until-verification on a single expiry; and the "checks saved" of §21's early settlement are revealed as a loan against the expirable ground: the settled verdict survives the death of its shortcut iff the saved checks were verified before the clock ran out — expiry-insurance, priced by the core.

The Epoch Boundary Theorem (`lean/EpochBoundary.lean`, empty axiom list, structural). Call a verdict *epoch-blind* at a marking if it is invariant along every finite chain of verify and expire events — i.e. if the protocol refuses to distinguish epistemic refinement from validity change. Then, within this model (markings over {T, F, Z}, the greedy evaluation, the two event kinds):

> a verdict is epoch-blind **iff** it is constant over all markings whatsoever — iff the assertion reads none of its grounds.

Contentful conclusions cannot survive unrestricted epoch crossing; the empirical census (2,906 formulas, 0 contentful survivors) is thereby upgraded from evidence to a theorem for every formula of the language. A separation witness (`epochs_matter`) completes the picture: a hereditary verdict destroyed by one expiry — so intra-epoch warranty and cross-epoch persistence are provably different notions, and the epoch boundary is a logical necessity, not an administrative convenience.

The institutional reading. For an admission formula over grounding atoms the layer yields a three-coordinate cell — epistemic status (the global supervaluation of §17: Z is *not established* and is never conflated with falsity), operational decision (the greedy verdict: default deny), and warranty grade — and an event ledger in which every verification tick carries its source. The closed-world loan — "no proof of revocation, hence not revoked" — cannot be taken inside the logic at all: $\neg R$ at $R = Z$ evaluates F/until-verification, an argument from absence never yields T. It can enter only as a tick without a source, which the ledger exposes as an *ungrounded verification event*. A worked, frozen artifact (admission condition; expiry branch; revocation branch; the rejected ungrounded tick) ships in the repository (`vrg/epoch_artifact.py`, deterministic JSON ledger).

23. The price of derivations: transport, not creation (MEASURED + Lean)

The price list of §3.1 concerns laws; it extends to *paths*. Take the 12 alive entailment rules of §3.2 as the only links, the 2 fallen ones (¬¬-elimination and tautology-in-conclusion) as a loan library, and close premise sets under chains, bounded to a 153-formula pool over three atoms; every closure is cross-checked against the semantic entailment of §3.2 (0 violations — the chains never lie; `zderive.py`). Three facts emerge.

From nothing, nothing — even on credit. The closure of the empty premise set is empty, with and without the loans: the battery has no axiom rule. And yet ZTL-tautologies *exist* — the guarded forms $\neg q \rightarrow \neg q$ and $\neg(p \oplus p)$ are true under every assignment, because denial is classical (¬Z = F; §3.1's "a denial is free") — and the battery cannot mint even those. The alive rules are *transport, not creation*: classical logic mints truth from form; ZTL's free truths must *enter*, as verified premises.

The one-way street, priced. From {p} the ladder ¬¬p is earned (¬¬-introduction is alive); from {¬¬p} the alive closure is {¬¬p} itself, and the ¬¬-elimination loan unlocks fourteen formulas — the first measured *on-credit derivations*, p among them. What classical logic cannot even see (the step is invisible inside it) is here a priced borrowing with a named creditor.

The gap, honestly. The rules are incomplete for the semantic entailment even on the small pool ({p}: 24 entailed, 15 derivable) — part pool-boundedness, part missing law-rewrite links, part genuine; the split needs a bigger pool, not a claim.

Kernel-checked (`ZNoAxiom.lean`, empty axiom list), and in the general form the 153-formula closure could only sample. Two halves that say something only together.

The tautologies exist under EVERY valuation, marks included. `¬q → ¬q` and `¬(p ⊕ p)` are T for every assignment, not merely across a pool — this is where the greedy denial earns its keep, since `¬Z = F` is classical and no mark can drag a formula built on denials down.

And nothing is derivable from nothing — for ANY calculus whose every rule demands a premise. Not "these twelve rules happen not to derive it": no such system can, whatever its rules are. Together: **there is a formula true under every valuation that no premise-requiring calculus derives from the empty set.** Truth of that shape has to be brought in, not minted — which is what "transport, not creation" says.

What is hypothesis and what is conclusion. That the twelve alive rules of §3.2 all have non-empty premise sets is a fact about the battery, recorded and measured there; it is the HYPOTHESIS of the theorem, not its conclusion. What is proved is that the hypothesis suffices, for every rule set at once. The one-way street and the incompleteness gap in this section remain measured.

24. Identity: a = predicate on credit (MEASURED + Lean)

The first-order layer of §6 had quantifiers but no identity. We add it, and the zero-trust principle turns the classical axioms of identity — reflexivity and Leibniz's substitutivity — from free laws into *earned verdicts*. The reading is the native one: a domain individual is a *reference*, and a **marked** individual is one whose reference is not yet verified — the individual-level face of Z (§10), a null pointer, an unresolved description, `1/0`. Equality inherits the mark:

```
Eq(a, b) = T   if a, b are grounded and denote the same object,
         = F   if a, b are grounded and denote different objects,
         = Z   if a or b is a marked (unverified) reference.
```

Four facts, all measured total in `zeq.py` over a domain of grounded and marked individuals, and all kernel-checked in Lean (`ZEq.lean`, nine theorems, **empty axiom list**) — on a fixed five-individual model (three grounded, two marked), as instance certificates rather than structural theorems over a class of models; the exhaustive statement is the measurement. The positive half is no longer instance-bound: `ZEqGeneric.lean` (§25) proves it for an arbitrary type with decidable equality; what stays exhaustive is the failures, which need only a countermodel. (Leibniz here is a one-line `congrArg`: once an equality is earned, = is genuine identity, so substitution is congruence — the content is in *when* it is earned, not in the substitution.)

- **Reflexivity is earned.** `Eq(a,a)` is T for a grounded reference but **Z** for a marked one — self-identity is not free; an unverified reference is not even certified equal to itself. This is the applied regularity R3 (§26), *identity is not earned*, now made a predicate: it is the same phenomenon as the uncertified `id` on a marked argument (§14), the non-registrable stream `x ≠ x` (§13), and `1/0 = 1/0 → Z` (§15). Reflexivity is therefore *not a law* (`refl_not_free`), the exact identity analogue of the fallen `¬¬p = p`.
- **The rule/law split (§3.2), on identity.** Symmetry holds as a rule (`Eq(a,b) ⊨ Eq(b,a)`, no violations) but the biconditional *law* `Eq(a,b) ↔ Eq(b,a)` fails wherever the equality is Z — because ↔ over Z is F (quarantine is detectable, `↔(Z,Z)=F`). Transitivity holds as a rule.

- **Leibniz substitutivity survives as a rule — salva veritate, not salva Z.** `Eq(a,b) ⊨ P(a) ⇔ P(b)` for every generic predicate P (zero violations of the licensed substitutions): an earned equality means the two references are literally the same object (`eq_forces_same`), so substitution is congruence (`leibniz_congr` is a one-line `congrArg`). Where a reference is unverified the equality is never T, so the substitution is simply never licensed — you cannot launder a mark through identity (`no_laundering`).
- **Identity of indiscernibles fails on the mark.** Two distinct marked references are indiscernible — Z under every generic predicate — yet $\mathrm{Eq}(m_1, m_2) = Z$, not T. Unverified references are not made equal by the mere absence of a distinguishing predicate.

So ZTL keeps the *consequence relation* of classical identity (Leibniz, symmetry, transitivity as rules) and loses exactly the free half — reflexivity-on-credit and indiscernibility-on-credit — by the same coin as everywhere else. And the marked individual is the seed of the next section: a non-denoting term is an unverified reference.

25. Free logic and definite descriptions: the mark as non-denotation (MEASURED + Lean)

Classical logic assumes every singular term denotes; *free logic* drops that assumption. Its schools disagree on what an atomic statement about a non-denoting term is worth — negative free logic says **false**, positive says some are **true**, supervaluational leaves a **gap** (but keeps excluded middle super-true), and the neutral school (Lehmann; weak-Kleene on non-denoting atoms) gives it a **third value**. ZTL's answer is the neutral one made operational — the atom takes the **mark** Z — with one decisive difference: the mark is **greedy**. It does not propagate as a gap and does not complete to a super-true middle; it evaporates at the first operator, so excluded middle on a non-denoting atom is **F**, not a gap and not super-true. Z was always the value of "unverified until verification"; a description not shown to denote is precisely that.

The spine is a bridge back to §24, and it is Quine's *"no entity without identity"* made literal:

```
E!(t)  :=  Eq(t, t)          — existence *is* earned self-identity.
```

A grounded reference is equal to itself, so it exists (E! = T); an unresolved reference is not, so E! = Z. Non-existence is therefore not asserted through E! — `¬E!(τ) = ¬Z = F`, always — but it is still sayable: a domain verified to be empty of φ earns `¬∃x.φ = T`, so "the F does not exist" is asserted Russell's way, as a denied existential. E! and ∃ part company on the negative side, and Russell's paraphrase is the honest way to deny. A definite description `ιx.φ(x)` — "the x such that φ" — denotes the unique grounded satisfier of φ, uniqueness decided by the = of §24; with zero or several satisfiers it is a marked reference. This is Russell's uniqueness, but failure lands on the mark, not on falsity.

Measured in `zdesc.py`, with the first-order core in Lean (`ZDesc.lean`, eight theorems, **empty axiom list**):

- **The sharp divergence: excluded middle.** For a non-denoting atom, `P(τ) ∨ ¬P(τ) = F` in ZTL, whereas supervaluational free logic makes it *super-true*. ZTL marks the gap rather than completing it; the mark is not a value, so the middle is not excluded (`lem_fails_nondenoting`).
- **Divergence from Russell.** "The present king of France is bald" is **F** on Russell's `∃x(Kx ∧ unique ∧ Bx)` and **Z** in ZTL: same sentence, Russell asserts falsity, ZTL refuses the assertion and marks it. (Compare §18, where Russell's *paradox* is contained rather than exploded — the same refusal to assert on unearned ground.)
- **Greedy propagation.** `P(τ)` alone is Z, but `P(τ) ∨ ⊤` is T (`mark_evaporates_in_compound`): a non-denoting term poisons only the atom, never a compound.
- **An internal witness.** `1/0 = ιx.(0·x = 1)` has no grounded x, so it is a marked reference and `1/0 = 1/0` is Z — dead-on with the arithmetic of §15. The construction reaches the same verdict the number line did.
- **Free universal instantiation.** The classical law `∀xφ ⊨ φ(t)` FAILS for a non-denoting t (`∀xφ` is T over what exists, `φ(τ)` is Z, and `T → Z = F`); the repaired free-logic law `∀xφ, E!t ⊨ φ(t)` holds, with existence the earned self-identity of the spine. Quantifiers range over what exists.

The nearest ancestor is Kleene's partial logic (an undefined term takes the third value, motivated by partial recursive functions); ZTL's delta is that the mark is **greedy**, so a non-denoting term inside a tautology-shaped compound still collapses to a verdict rather than staying undefined. *Where this stands:* the identity-of-indiscernibles failure and the free instantiation law are now theorems in Lean on the empty axiom list (`ZIndisc.lean`, `ZFreeUI.lean`), alongside the ι operator, the existence bridge, and the excluded-middle divergence. Both were measurements until 2026-09-04; the gap was never the size of the domain but the order of the quantifier — indiscernibility ranges over predicates, and a finite named list of them keeps the check decidable where the function type would not be. *Where the boundary now runs:* a falsified law needs one countermodel and has it, so those failures hold generally. The positive half of identity is no longer domain-bound either — `ZEqGeneric.lean` proves it for an arbitrary type with decidable equality and an arbitrary choice of marked references: an earned equality is genuine identity, substitution through it is congruence, equality is never earned through a mark, reflexivity holds exactly on the verified, and indiscernibility forces identity among grounded references. The positive half of the instantiation schema went the same way the next day — `ZFreeUIGeneric.lean` proves it for an arbitrary type with decidable equality, an arbitrary marking, an arbitrary V-valued predicate and an arbitrary range: wherever the premise $\forall^G \psi \wedge E!t$ is earned so is $\psi(t)$; on a marked term the premise is never designated, so the schema licenses nothing there; the repair is not vacuous; and an empty range is no loophole, since the refusal comes from the mark rather than from the range being inhabited. *What is still finite is the RANGE, not the domain*: a strict universal has to be computed, and computing one over an unsurveyable range would be exactly the survey this logic refuses to call an act. What remains proved by exhaustion is now only the failures, and necessarily so: over an arbitrary domain they are false, since a domain without marks is classical and there the laws hold. A countermodel belongs in a witness, not in a general theorem.

The indefinite description, and choice as an act. The definite ι has a companion: Hilbert's ε, the choice term $\varepsilon x.\psi$ — "*an* x such that φ". Where ι demands uniqueness, ε demands only a witness and picks a canonical one; where none exists, the choice is unearned and $\varepsilon x.\psi$ is a marked reference. Three facts (measured in `zeps.py`, core in `ZEps.lean` on the empty axiom list). First, the **ε–∃ bridge**: $E!(\varepsilon x.\psi) = T$ exactly when $\exists x.\psi$ — the choice term denotes precisely when the existential is earned, and in ZTL the existential is itself the strict-T witness of §6, so the two coincide. Second, ι and ε **divide the labour of reference**: on a uniquely-satisfied φ they agree, but on a multiply-satisfied one $\iota x.\psi$ is marked (no unique referent) while $\varepsilon x.\psi$ denotes a choice. Third, **the empty choice is the mark** — $\varepsilon x.(\bot)$ earns Z, never a free F: to choose from nothing is not an act. This is the operational reading of the Hilbert operator — a choice is an act, and only an act grounds a term — the same discipline the whole logic runs on, now at the level of singular terms.

Note (v1.4): three kinds of non-denoting are marked alike, and that is the right answer at the level of value. A description can fail to denote in three ways — no satisfier, several satisfiers, or a satisfier whose denotation is simply unverified — and `zdesc` gives all three the same mark. The Lean theorems say so directly: `iota_empty_marked` and `iota_multiple_marked` both return Z, and they are right to. Both are non-denoting, neither earns existence, and *as a value* nothing separates them. The distinction between the three is real, but it does not live in the calculus.

We looked, because it seemed as though it should. Measured 2026-08-19 against the published code, imported and unmodified: **no verdict changes anywhere.** `E` is a typed evaluation event, not a value; it does not enter connectives; every compound answers exactly as before. So the honest headline is the negative one — *the distinction buys no logical power at all*, and nothing in the published semantics of this section needs correcting.

Where it does live is one floor up, in the work order. The judge does not only answer; it names the next check. In all three cases it named the same one, and only one of the three can be carried out. With no satisfier there is nothing to examine and never will be. With several satisfiers examining does not help either, for a different reason: the world is not missing anything, the claim is, and what is owed is a *stipulation* rather than a check — the corpus already had the word and was not using it here (`zpassport`: UNDERDETERMINED, several classical models fit; stipulate one and the claim grounds). Only where the denotation is merely unverified is "go and check" a real instruction. Worse, the first two were marked OPEN — a status asserting the question can be opened.

That is a defect of instruction, not of verdict, and the two are different objects. It was invisible from inside the calculus precisely because the calculus was behaving. The repair is in the disposition layer: a fifth disposition (the judge's dispositions are the kinds of work order it can issue) for "this ground has no subject", a rule that no order is issued which is already known to be unfillable, and — since a declaration of absence is a premise rather than a discovery — a bill: on declaring that a ground has no subject, the judge names which settlement that declaration foreclosed. Where the ground would have decided the matter, the bill is heavy and the declaration becomes contestable on the world; where the matter stands on other grounds, the bill is empty. A missing subject halts its own predicate, never the proceeding. None of this is an erratum against the published record, and it is recorded here rather than quietly folded in, because the temptation to inflate a correct verdict with a wrong instruction into a semantic correction is exactly the kind of over-claim this paper spends its caveats resisting.

26. Cross-cutting regularities

Three facts recurred in every applied chapter; we fix them once.

R1. Two levels of equality. On every floor (sets, functions, arithmetic) the operations coincide at the representation level (the machine sees equality), yet verdict-equality refuses: the very act of recognizing identity is default deny. The engineering level serves the solver, the verdict level serves assertions.

R2. Two-registeredness reproduces itself. The verdict and solver variants of each construction (preimages, cardinalities, probabilities) arise without special design — as consequences of the generating principle and the §9 argument for the necessity of two registers.

R3. Apartness is earned, identity is not. The difference of two unverified objects is earned by a finite witness (diverged intervals, diverged prefixes); identity is earned by nothing short of full verification. Hence in one stroke: {Z,Z} ≠ {Z}, m−m ≠ 0, the non-registrability of streams (§13), and the NaN signature x ≠ x.

27. Roadmap

~~a syntactic cut-elimination procedure with complexity bounds~~ — DONE, E55 (`ZCutElim.lean`, §5): the procedure, its bound `B0` as a function, the classical pair refused on atoms and granted on compounds; what remains open there is whether the bound can be tightened — measured loose by five orders on the pools, and a tightening would need the hard instances the pools lack; the mining of the equivalent quasivariety scouted in §3.7 (axiomatization, subquasivariety lattice, a representation theorem replacing Plonka sums — a separate work); a cheap characterization of the hereditary warranty grade of §19 (sound is one pass over completions; is hereditary computable without enumerating refinements? — MEASURED narrowing, 2026-07-12/13: the fence depth is exactly m−1 in the number of marks — sufficient for every sound verdict (violations cannot hide in full completions) and necessary by the guard family (b_1∧…∧b_{m−1}) → (a→a), checked at m = 2,3,4,5 (`zverify` §§5–6); hence NO constant-depth characterization exists — and E57 (`ZHeredTaut.lean`, §19) closes the remaining question: no structural, non-enumerative criterion exists either, unless P = coNP, because heredity of the guarded witness $(\wedge_i\ x_i \vee \neg x_i) \to \psi$ is exactly tautology-hood of ψ — and E58 (`ZWidthHard.lean`) does the same for the width: n+1 iff ¬ψ is satisfiable, so the exact width is NP-hard and `joint`'s width-1 cut is the right one — and E59 (`ZReceiptHard.lean`, §19) does the same for the receipt: a named atom is idle exactly when its sibling is unsatisfiable, so the exact receipt is NP-hard and `labF`'s candidate list is the right cut); A Lean port of the parameter (arbitrary-domain) tableaux of §6 — BEGUN, and the two pieces done are the ones that decide what the rest costs. `ZParamSound.lean` measures the tier of the four rules (see §6): three are sound on the empty axiom list and the fourth is not. `ZParamSyntax.lean` lays the ground BOTH quantifier rules stand on — the monadic syntax with parameters, satisfaction as a RELATION rather than a computation (forced: the greedy ∀ over an arbitrary domain is not decidable), and the two lemmas without which neither rule is licensed. FRESHNESS: reassigning a parameter that occurs nowhere in a formula leaves every verdict of it unchanged, quantifiers included — this is what lets δ name a witness. INSTANTIATION: putting a parameter into a formula is the same as putting its value on the evaluation stack — this is what lets γ and δ instantiate at all. Every soundness argument for these rules is those two plus bookkeeping. (The instantiation lemma REPLACES a stack position rather than inserting one, because `inst` does not shift the other indices; written the other way the two sides disagree above the substituted depth, and the error would surface only in the tableau, files later.) The RULE STEPS are then proved on branches (`ZParamTableau.lean`): γ on T:∀ instantiates with any parameter already in play and the model does not move at all; δ on T:∃ points a FRESH parameter at the witness, and that this disturbs nothing else on the branch is exactly the freshness lemma earning its place. Both preserve satisfiability, which is what a sound step has to do. The third, γ on F:∃, is stated with an explicit TOTALITY hypothesis about the model rather than proved outright: it needs a value for the instance, and over an arbitrary domain the existence of a value for a quantified formula is itself classical. The hypothesis is visible in the statement instead of hidden in the definition of satisfaction, where every other theorem would have paid for it — the same split measured in `ZParamSound`, reappearing one level up. CLOSURE is proved too (`ZParamClosure.lean`): a closed branch — two clashing signs on one formula — has no model. That needed a lemma which is not a definition once satisfaction is a relation: A FORMULA HAS ONE VALUE. Determinism is proved constructively, and the quantifier case is where it could have failed — two admissible values could differ only if one were T and the other F, and then the T one forces the universal, which forces the other to T as well. The two specifications are played against each other rather than deciding the undecidable universal.

The PROPOSITIONAL steps are carried over to the quantified language too (`ZParamProp.lean`), each read off a cover lemma of §5 rather than invented, which is why the weak signs appear exactly where the tables put them: `F:¬φ` yields P and not F, because `¬v = F` holds at T AND at Z, and writing F there would claim more than the table allows. Branching and non-branching steps are different theorems — a non-branching step says the extended branch is satisfied, a branching one says AT LEAST ONE successor is, which is what makes closure of every branch a proof.

So the soundness argument for the quantified calculus is complete IN ITS PARTS, and now for the whole language rather than its quantifier fragment: every rule carries satisfiability forward, closure denies it. THE SEARCH IS NOW BUILT AND PROVED SOUND (`ZParamEngine.lean`, E48), on the layer E47 found missing: nodes carry a TAG from a four-element inductive (`t f p n`) instead of a function, a translation sends tags to signs, formulas have decidable equality, and `closedB` decides closure — with `closedB_sound` proving that what it calls closed is closed in the sense above, hence has no model. The four tags are not four values: ZTL is two-valued with a mark, and a tag says which VERDICTS a node admits (`n` = "not T": F, or the still unanswered Z); P and N do not clash precisely because both admit the mark. The search runs a fuel-bounded worklist: a closed branch is discharged, otherwise the first node with a rule that adds something new is expanded and its successors return to the list; running out of fuel claims nothing, as §6's undecidability requires. It applies EXACTLY the rules proved sound on the empty axiom list — the eight propositional steps, γ on T:∀ and F:∃, δ on T:∃ — and F:∀, the rule that needs `¬∀ → ∃¬`, is deliberately absent. The end-to-end theorems: `search_sound` (a run that closes every branch shows no initial branch has a model, under any assignment, for a total interpretation) and `entails_of_closed` (a closed run on Γ ⊢ φ, premises tagged t and the conclusion n, proves that every total model making each premise T makes φ T — constructively: the conclusion's value is PRODUCED by totality and shown to be T, not obtained by refuting its negation). Three runs are kernel-evaluated: ∀xP(x) ⊢ P(c) closes, P(c)∧Q(c) ⊢ Q(c) closes, and ¬∀xP(x) ⊢ ∃x¬P(x) — the fallen bridge — returns `stuck`, not `closed`. The fresh parameter is a theorem, not a side condition: it is a SUM of the indices on the branch, and `freshFor_fresh` proves it occurs nowhere. E51, the same day, gave the atoms any number of places (`atom : Nat → List Trm`, with `trmVals_fresh` / `trmVals_inst` carrying the two load-bearing lemmas over argument lists) and the two rules the first search lacked: in the greedy register EVERY COMPOUND ANSWERS T OR F (`compound_two_valued` — `evalF_classical` for the parameter language), so N on a compound is F and P is T, and only atoms keep a weak sign; before that a compound conclusion had no rule and `P(c) ⊢ P(c) ∨ Q(c)` was `stuck`. Three more runs close by the kernel — `P(c) ⊢ P(c) ∨ Q(c)`, `∀x∀y R(x,y) ⊢ R(a,b)`, `∃x R(x,x) ⊢ ∃x∃y R(x,y)` — and the VALID swap `∃x∀y R(x,y) ⊢ ∀y∃x R(x,y)` returns `stuck` for the honest reason: its conclusion N:∀ promotes to F:∀, the one rule this search does not have, because it is the classical one — the split of `ZParamSound`, now visible in a whole run. Two repairs came from the kernel runs and not from reading: γ's candidates are the parameters in play (a bound growing with the branch made γ spawn instances forever), and δ fires once (it re-fired with a fresh parameter every round). E54 (2026-09-06) turned to completeness and found, before proving anything, that the search as shipped could not REPORT an open branch: probed on invalid sequents — which none of the runs above did — `P∨Q ⊢ P` returned `noFuel` at fuel 10, 40 and 200 (a branching rule fired whenever ONE successor was absent, so on the open successor it re-fired forever), and the valid `∀x∃y R(x,y), P∧Q ⊢ Q` returned `noFuel` with the generator first among the premises and `closed` with it last (new nodes went to the head and `pick` scanned from the head; `sat_rotate` of E47 was proved for exactly this and never wired). Neither touched soundness, which speaks only of `closed`; both made the Hintikka half unreachable. Both are fixed and kernel-checked (`run_or_elim_open`, `run_generator_first`). δ_2 then entered BEHIND A FLAG: `search false` is the engine above, `search true` adds `F:∀xφ → N:φ(c*)`; its soundness enters the corpus as the hypothesis `Delta2Step`, discharged classically OUTSIDE it (`inventory/probes/delta2_step_classical.lean`, measured by `ПАРАМЕТР-ЯРУС.py`) — the corpus keeps its invariant, the swap closes (`run_swap_with_delta2`), and the fallen bridge stays `stuck` even with δ_2, since N against P is not a clash. What remains of the port is the INFINITE half of completeness, stated in §6 with its parts named; the finite half is `ZParamHintikka.lean`; a fragment-embedding theorem for the remaining three traditions of §1 (as originally listed) — ALL SIX are now done, and the entry stays as the record of what it asked for and what was delivered against it. The first twin — IEEE 754's NaN, the first tradition of §1 — is embedded in `ZNaN.lean` (E49, empty axiom list): the fragment as the standard states it (data, the four mutually exclusive relations of §5.11, the predicates of Table 5.1, the §6.2.3 propagation rule); their Boolean layer proved classical (!= is the complement of == on every relation, the unordered one included); infection proved to be the lazy register (`emb` is a homomorphism for + and −); every ordered predicate proved to be `SignT` of a ZTL atom (<= and >= of the GREEDY disjunction — `zor Z Z = F`, and IEEE's <= is false on a NaN too); the unordered predicate proved to be `SignT (isZ ...)` — quarantine detectable from inside, §3.4, under IEEE's name; and != proved to be `SignN` of the equality atom and NOT the T-sign of the greedily negated one: off the mark the two coincide, on the mark IEEE affirms `x != y` while ZTL's `¬(x = y)` is `¬Z = F`. So the NaN signature "not equal to itself" is shared on `x == x` and splits on its complement — one refusal and an affirmation against two refusals (§18). The arithmetic boundary sits in the same file: `0 × NaN = NaN` against the earned `0 · mark = 0` of §15, two `rfl`s. Not modelled: rounding, signed zero, infinities, the signaling NaN — one algebraic core, not the standard. Every `Int` order lemma measured carries propext, so both sides read the order off one three-way comparison and no such lemma is used. The second twin — SQL's NULL — is embedded in `ZNull.lean` (E50, empty axiom list): the truth values TRUE/FALSE/UNKNOWN with the standard's three tables, the comparison predicates, WHERE, CHECK and the <boolean test>. Proved: the expression layer is a HOMOMORPHISM onto the lazy register `knot/kand/kor`, cell for cell — and not onto the greedy one (`NOT NOT UNKNOWN` is UNKNOWN, `¬¬Z` is T); the comparison with NULL is the SAME mark atom IEEE's == landed on — two traditions, one ZTL atom; the boundaries are the four signs — WHERE keeps iff `SignT`, CHECK passes iff `SignP` ("satisfied iff not False": UNKNOWN passes), IS TRUE / IS FALSE / IS NOT TRUE / IS NOT FALSE are `SignT / SignF / SignN / SignP`, IS UNKNOWN is the mark test `isZ`. So SQL runs TWO boundaries with opposite defaults — `x < NULL` and `x >= NULL` are both dropped by WHERE (§4's "false in both polarities") and both PASS a CHECK — while ZTL's collapse is WHERE's alone: the mark falls to F, never to T (`collapse_is_where`). At the WHERE boundary SQL and ZTL's greedy verdict AGREE on every negation-normal search condition (`where_agrees_nnf`: the two registers share their T-cells at every connective) and part exactly where a negation stands over a compound — `¬¬Z`, the signature cell; at CHECK they part already on `TRUE AND NULL`. And SQL's two equalities: `NULL = NULL` is UNKNOWN while `NULL IS NOT DISTINCT FROM NULL` is TRUE (the rule DISTINCT and GROUP BY run on) — ZTL has one equality and it withholds (§12's "inconsistency not inherited", as `two_equalities`). Not modelled: tables, joins, aggregates — the truth-value layer and its boundaries, not the language. Fifth core. The third twin — taint tracking / information flow — is embedded in `ZFlow.lean` (E52, empty axiom list), with Denning's lattice (1976) in its two-point fragment as the COMMON TARGET: its join proved a lattice join; Perl's taint propagation (perlsec) proved to be that join, and so are ZTL's pointwise functions on pedigreed elements (E7's `taint`) and `+ −` on marked integers; Denning's flow check into a low object is Perl's sink and is `SignT` of ZTL's "verified" atom. Where the three part is DECLASSIFICATION, and each position is a theorem: Denning's rule never lowers a class (`join_absorbs_H`); Perl lowers it BY CONVENTION — `regexCapture` clears the bit while its payload is the tainted input's, a flow the lattice forbids (`perl_launders`); ZTL lowers it BY PROOF or not at all — no chain of pointwise functions clears a mark (`ztl_no_laundering`, E40 restated in the lattice), and the one clean value × returns from a marked operand, `0 · mark = 0`, is the value forced for every reading (`zero_forced`), where Denning's syntactic join still says H (`denning_overtaints_forced`): dependence in the syntax against independence in the value. Not modelled: multi-level lattices, implicit flows through control, TaintDroid's sources and sinks. Sixth core — and with it the §1 claim that six traditions implement fragments of one logic rests on six embedding theorems rather than six worked cases. The fifth twin went the same day: `ZDempster.lean` formalises Dempster–Shafer as they define it (focal elements, Bel, Pl) and proves our verdict to be their {0,1}-threshold in all three cells, for every finite frame and every proper mass assignment — see §16, including the properness condition the first draft got wrong. `ZProv.lean` (empty axiom list) formalises the ALGEBRA of semiring provenance — the free commutative semiring on sources, + for an alternative derivation, · for a joint requirement — and maps it into ZTL: evaluation is a homomorphism into the LAZY register, which satisfies every semiring law; withdrawal of a source never raises trust along any derivation; a joint requirement dies with any one source while an alternative survives; and on mark-free trust the two registers coincide, so the Boolean case the 2007 paper starts from is recovered exactly. The sharp half is negative and belongs to us: the GREEDY operations carry no semiring structure at all — no constant whatever is neutral for `zor`, none for `zand`, the obstruction being the mark (`zor F Z = F`, `zand T Z = F`). So an algebra of trust in derivations lives in the lazy register, and the verdict register is where it is CASHED — cashing being provably not a homomorphism. That is why the two registers had to be two. What is NOT formalised is their database semantics: K-relations and the annotated relational operators. So one tradition's algebraic core is mapped, not the tradition; a practical zero-trust validation library (verdicts with warranties + evidence combination + provenance); temporal operators over the verification tree (□/◇/until against the grade semantics of §21) once a concrete user forces them, and the grade automaton at scale; The interactive studio — natural language negotiated into ZFL and judged by the measured engines — already ships in the repository (`tool/`); a possible essay, "Reinventing Bochvar through NaN", remains on the horizon.

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Acknowledgements and AI disclosure

This work was carried out with the substantial participation of the AI system Claude (Anthropic) in a dialogue setting: the system generated the text, the test-bench code, and the Lean proofs. Across versions the dialogue ran on four Claude models, and the attribution is kept honest, read from the commit trailers: Claude Opus 4.8 — the original corpus (v1.0), the 2026-07-14/15 additions (the three-laws capstone §3.1, the Finn attribution and reconciliation in §3.8, the paradox-engine synthesis opening §11, the Łukasiewicz pedigree and the genetic order in §4), the kernel-checked theorems of §§13, 14 and 17 in `ZExped.lean`, and the v1.3 assembly and PDF build, applying a fresh-eyes review by Claude Fable 5; Claude Fable 5 — the v1.1 same-day correction and the v1.2 assembly (the census of sixteen and its Lean clone equalities, the fence-depth theorem, the warranty ladder at scale, the naming of the lift, the temporal layer §§21–23 with `ZTime.lean`

and EpochBoundary.lean, the expiry and derivation stands, and the v1.2 PDF build), the kernel-checked facts of §§11 and 12 (Facts.lean, ZSets.lean), ZNum.lean, and the warrant-grade case studies of the corpus (Plato_Equality.lean, Plato_Conservativity.lean, Hume_Guillotine.lean); Claude Opus 5 — the first wave of v1.4, from the fence-depth theorem for every m to the foundation of the parameter-tableau port: ZFenceDepth.lean; the embeddings ZProv.lean, ZDempster.lean, ZYablo.lean and ZAbsInt.lean; ZTaint.lean, ZCombine.lean, ZIgnorance.lean and ZNoAxiom.lean; ZParamSound.lean, ZParamSyntax.lean, ZParamTableau.lean, ZParamClosure.lean and ZParamProp.lean; the receipt results of §19 with Receipt.lean, Linear.lean and LabelExact.lean; NoGift.lean, ContextClosure.lean, ClassicalAgreement.lean, ZIndisc.lean, ZFreeUI.lean, ZEqGeneric.lean, ZFreeUIGeneric.lean and ZNumCoherent.lean; the reliance bridge (RelianceBridge.lean, Layering.lean); the position against Tomova's criterion in §4; and the accompanying benches; Claude Fable 5.1 — the search and the last three embeddings (ZParamEngine.lean, ZNaN.lean, ZNull.lean, ZFlow.lean), the finite half of completeness (ZParamHintikka.lean), cut elimination (ZCutElim.lean), the three hardness results (ZHeredTaut.lean, ZWidthHard.lean, ZReceiptHard.lean), LabelExactDefinite.lean, ZReconverge.lean, ZNumPrice.lean, the claim-grade correction of the complexity statements, and the v1.4 assembly. All design decisions, fork choices, hypotheses, and the final responsibility for the content rest with the human author. In accordance with COPE/ICMJE recommendations, the AI system is not listed as an author. The reliability of the results does not depend on trusting the AI: every numerical claim is checkable by the repository code (run_all.py — full regression), and every Lean claim by the Lean 4 kernel (empty axiom list, #print axioms).